QUALIFICATION STANDARD AND PLAN (QSP)

Weapons Engineering Technician (W Eng Tech)
NQual (AKQR)

MITE ID
Residential Phase – 118944

Regular Force: Weapons Engineering Technician 00366 (W Eng Tech)
Training Authority (TA): Commander Naval Training Group (CNTG)
Curriculum Control Authority (CCA): NFS(P)

MARPAC RDIMS #1417022
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RECORD OF AMENDMENTS

1. The following versions of this QSP have been approved for use in support of the conduct of training. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.0	QSP creation	21 Feb 2011	DMTE 2-4-2
1.1		22 Aug 2013	O.S.B.//Cdr Benoit CFNES Cmdt
1.2		Jul 2014	Cdr D.J. Benoit CFNES Cmdt
1.3	Addition of Torpedo Pos	03 Jun 2024	NTG HQ
1.4	Administrative Correction made t o1.2 QS amend # 2024 - 148	14 Jun 2024	NTG HQ
1.5	Addition of Chapter 3 and 4 to 1.3 TP amend# 2024 - 75	27 Jun 2024	CCA
1.6	QSP change to new format Addition of Unit Technical Inspector MITE 302486	29 Aug 24	NTG HQ

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the NTG Website. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Training Group (CNTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the following:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-002, Non-Commission Member General Specification published 09 February 2011;

   c. and A-PD-055-002/PP-001, Weapons Engineering Technician 00366 published 01 September 2011;

3. This QSP publication is effective on receipt and supersedes AKQR QSP revision 1.2, dated July 14 (RDIMS #1417022). This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS at Chapters 1 and 2 were developed by the Qualification Standard Writing Board (QSWB) convened 15 Jan 2023 at NFS(P) working in collaboration with subject matter experts drawn from participating organizations. The subsequent TP at Chapters 3 and 4 were developed by the Training Plan Writing Board (TPWB) convened Apr 2023 at NFS(P) working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a QSP Deviation Request is required.

QUALIFICATION STANDARD

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CHAPTER 1 QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the requirements and training for the Weapons Engineering Technician (W Eng Tech) to perform their duties as a Sonar Maintainer. Possession of this qualification will allow personnel to operate, troubleshoot, and perform preventative and corrective maintenance on sonar equipment.

GRANTING OF NQUAL

2. A trainee will achieve the qualification (AKQR) W Eng Tech Sonar Applications when the standards listed in the Performance Objectives (POs) have been met.

a. Sonar Applications, MITE 118944

CO-QUALIFICATION

3. In addition to the performance objectives specific to this training, the following additional qualifications are included in this training program to meet the requirements of the job. The MITE and NQuals associated with these qualifications shall be granted upon completion of the training program and shall be listed within the course report and will be delivered IAW the following:

a. DRMIS, MITE 119097; and

b. Unit Technical Inspector, MITE 302486.

PREREQUISITES

4. To be selected for this course the loading authority must ensure that trainees meet the following prerequisites:

a. AKNV_ Jr W Eng Tech EQ2 MITE 118939 or MITE 123762; and

b. AKPK_W Eng RQS1 Academics MITE 118966.

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CHAPTER 2 PERFORMANCE OBJECTIVES

PO 001 Maintain Navigation Data Distribution System

1. Performance: Maintain the Navigation Data Distribution System (NDDS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to NDDS or training simulator

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Sonar Systems Maintainer shall maintain the NDDS IAW refs, by:

   a. Initializing NDDS, to include:

      (1) Powering-up;

      (2) Configuring the system to meet operational requirements; and

      (3) Verifying correct system operation.

   b. Performing CM on the Navigation Data Distribution System, to include:

      (1) Performing a functional check;

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(2) Identifying faults;

(3) Replacing/repairing the LRU;

(4) Verifying repair; and

(5) Completing technical administration.

4. Reference:

a. A01 – A07; and

b. B01 – B02.

5. Task Numbers:

a. A1 – A20.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 002 Maintain Inertial Navigation System

1. Performance: Maintain the Inertial Navigation System (INS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to INS or training simulator

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Sonar Systems Maintainer shall maintain the INS IAW refs, by:

   a. Initializing the INS, to include:

      (1) Powering-up;

      (2) Configuring the system to meet operational requirements; and

      (3) Verifying correct system operation.

   b. Performing CM on the INS, to include:

      (1) Performing a functional check;

      (2) Identifying faults;

(3) Replacing/repairing the LRU;

(4) Verifying repair; and

(5) Completing technical administration.

4. Reference:

a. A01 – A07; and

b. C01.

5. Task Numbers:

a. B1 – B17.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 003 Maintain Shipboard Integrated Navigation and Display System

1. Performance: Maintain Shipboard Integrated Navigation and Display System (SHINNADS).

2. Conditions:

a. Given:

(1) Supervision;

(2) References;

(3) Tools;

(4) Job aids;

(5) PPE;

(6) Assistance; and

(7) Access to SHINNADS or training simulator

b. Denied:

(1) Nil.

c. Environment:

(1) Any shipboard conditions at sea or alongside; and

(2) Day or night.

3. Standard: The W Eng Tech Sonar Systems Maintainer shall maintain the SHINNADS IAW refs, by:

a. Initializing the SHINNADS, to include:

(1) Powering-up;

(2) Configuring the system to meet operational requirements; and

(3) Verifying correct system operation.

b. Performing CM on the SHINNADS, to include:

(1) Performing a functional check;

(2) Identifying faults;

(3) Replacing/repairing the LRU;

(4) Verifying repair; and

(5) Completing technical administration.

4. Reference:

a. A01 – A07; and

b. D01.

5. Task Numbers:

a. C1 – C23.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 004 Maintain Sonobuoy Processing System

1. Performance: Maintain Sonobuoy Processing System (SPS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SPS or training simulator

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Sonar Systems Maintainer shall maintain the SPS IAW refs, by:

   a. Initializing the SPS, to include:

      (1) Powering-up;

      (2) Configuring the system to meet operational requirements; and

      (3) Verifying correct system operation.

   b. Performing CM on the SPS, to include:

      (1) Performing a functional check;

      (2) Identifying faults;

(3) Replacing/repairing the LRU;

(4) Verifying repair; and

(5) Completing technical administration.

4. Reference:

a. A01 – A07; and

b. E01 – E05.

5. Task Numbers:

a. D1 – D24.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 005 Maintain the Canadian Towed Array Sonar System

1. Performance: Maintain the Canadian Towed Array Sonar System (CANTASS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to CANTASS or training simulator

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Sonar Systems Maintainer shall maintain the CANTASS IAW refs, by:

   a. Initializing the CANTASS, to include:

      (1) Powering-up;

      (2) Configuring the system to meet operational requirements; and

      (3) Verifying correct system operation.

   b. Performing CM on the CANTASS, to include:

      (1) Performing a functional check;

      (2) Identifying faults;

(3) Replacing/repairing the LRU;

(4) Verifying repair; and

(5) Completing technical administration.

4. Reference:

a. A01 – A07; and

b. F01 – F17.

5. Task Numbers:

a. E1 – E72.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 006 Maintain Hull Mounted Sonar

1. Performance: Maintain Hull Mounted Sonar (HMS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to HMS or training simulator

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Sonar Systems Maintainer shall maintain the HMS System IAW refs, by:

   a. Initializing the HMS System, to include:

      (1) Powering-up;

      (2) Configuring the system to meet operational requirements; and

      (3) Verifying correct system operation.

   b. Performing CM on the HMS System, to include:

      (1) Performing a functional check;

      (2) Identifying faults;

(3) Replacing/repairing the LRU;

(4) Verifying repair; and

(5) Completing technical administration.

4. Reference:

a. A01 – A07; and

b. G01 – G06.

5. Task Numbers:

a. F1 – F12.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 007 Maintain Torpedo Decoy System

1. Performance: Maintain Torpedo Decoy System (TDS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to TDS or training simulator

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Sonar Systems Maintainer shall maintain the TDS IAW refs, by:

   a. Initializing the TDS, to include:

      (1) Powering-up;

      (2) Configuring the system to meet operational requirements; and

      (3) Verifying correct system operation.

   b. Performing CM on the TDS, to include:

      (1) Performing a functional check;

      (2) Identifying faults;

(3) Replacing/repairing the LRU;

(4) Verifying repair; and

(5) Completing technical administration.

4. Reference:

a. A01 – A07; and

b. H01 – H03.

5. Task Numbers:

a. G1 – G20.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 008 Maintain the Shipboard Electro Optic Surveillance System

1. Performance: Maintain the Shipboard Electro Optic Surveillance System (SEOSS).

2. Conditions:

a. Given:

(1) Supervision;

(2) References;

(3) Tools;

(4) Job aids;

(5) PPE;

(6) Assistance; and

(7) Access to SEOSS or training simulator

b. Denied:

(1) Nil.

c. Environment:

(1) Any shipboard conditions at sea or alongside; and

(2) Day or night.

3. Standard: The W Eng Tech Sonar Systems Maintainer shall maintain the SEOSS IAW refs, by:

a. Initializing the SEOSS, to include:

(1) Powering-up;

(2) Configuring the system to meet operational requirements; and

(3) Verifying correct system operation.

b. Performing CM on the SEOSS, to include:

(1) Performing a functional check;

(2) Identifying faults;

(3) Replacing/repairing the LRU;

(4) Verifying repair; and

(5) Completing technical administration.

4. Reference:

a. A01 – A07; and

b. I01 – I02.

5. Task Numbers:

a. H1 – H18.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 009 Maintain Combat Management System 300

1. Performance: Maintain Combat Management System 330 (CMS 330).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to CMS or training simulator

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Sonar Systems Maintainer shall maintain the CMS 330 IAW refs, by:

   a. Initializing the CMS 330, to include:

      (1) Powering-up;

      (2) Configuring the system to meet operational requirements; and

      (3) Verifying correct system operation.

   b. Performing CM on the CMS 330, to include:

      (1) Performing a functional check;

      (2) Identifying faults;

(3) Replacing/repairing the LRU;

(4) Verifying repair; and

(5) Completing technical administration.

4. Reference:

a. A01 – A07; and

b. J01 – J06.

5. Task Numbers:

a. J1 – J9.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Nil.

PO 010 Maintain Torpedo Handling System

1. Performance: Maintain Torpedo Handling System (THS).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to THS or training simulator

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Sonar Systems Maintainer shall maintain the operability of the THS IAW refs, by:

   a. Initializing the THS, to include:

      (1) Powering-up;

      (2) Performing system functional checks; and

      (3) Embarking/disembarking a torpedo, to include:

         (a) Identifying types of torpedos; and

         (b) Practicing torpedo handling system.

   b. Performing CM on the THS, to include:

(1) Identifying deficiency;

(2) Performing diagnostics;

(3) Conducting fault finding;

(4) Identifying failed LRU;

(5) Replacing LRU;

(6) Performing functional check; and

(7) Completing maintenance administration.

4. Reference:

a. A01 – A07; and

b. K01 – K05.

5. Task Numbers:

a. Nil.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Moving and securing the torpedo utilizes a multitude of overlapping steps. Any movement will finish with securing torpedo and equipment.

PO 011 Maintain Surface Vessel Torpedo Tube

1. Performance: Maintain Surface Vessel Torpedo Tube (SVTT).

2. Conditions:

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SVTT or training simulator

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Any shipboard conditions at sea or alongside; and

      (2) Day or night.

3. Standard: The W Eng Tech Sonar Systems Maintainer shall maintain operability of SVTT IAW refs, by:

   a. Operating the SVTT, to include:

      (1) Conducting flash-up;

      (2) Performing system functional checks;

      (3) Loading/unloading a torpedo IAW refs;

      (4) Directing a Torpedo Handling Crew; and

      (5) Test-firing a Torpedo in all modes IAW refs.

   b. Performing CM on the SVTT, to include:

(1) Identifying deficiency;

(2) Performing diagnostics;

(3) Conducting fault finding;

(4) Identifying failed LRU;

(5) Replacing LRU;

(6) Performing functional check; and

(7) Completing maintenance administration.

4. Reference:

a. A01 – A07; and

b. L01 – L03.

5. Task Numbers:

a. Nil.

6. Training Limitations:

a. Nil.

7. Remarks:

a. Moving and securing the torpedo utilizes a multitude of overlapping steps. Any movement will finish with securing torpedo and equipment.

TRAINING PLAN

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CHAPTER 3 TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. Lecture
   b. Demonstration
   c. Practice

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 92 training days.

Actual Training Time (ATT)	579 periods
Training Administrative Time (TAT)	0.00
Base Administrative Time (BAT)	0.00
Training Travel Time (TTT)	0.00
Course Duration (total calendar days) (ATT+TAT+BAT+TTT)/8	**92 training days**

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training.

Course Director				
Rank	MOSID	Qualifications Required	Qty	PO / EO

Instructors				
Rank	MOSID	Qualifications Required	Qty	PO / EO
MS/PO2	00366	W Eng Sonar Applications BIT	2	
W Eng Tech Manager	00366	W Eng Tech Maintenance Manager	1	

Guest Lecturers/Others (May be civilian)				
Lecturer	# Instructor	Subject	EO #	# period

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TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below:

a. Limitation 1; and

b. Limitation 2.

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

a. Optimum – 1:6

b. Maximum Load – 1:8; and

c. Minimum – 1:2.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2).

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

a. The Career Manager for career courses; or

b. NFS Campus Support for occupation and specialty courses.

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully complete all POs to achieve the qualification.

9. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500-22. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

ENABLING AND PERFORMANCE CHECKS

10. All Performance Checks (PCs) are based on Performance Objectives (POs) in Chapter 2 of this QSP and the standard listed must be met to achieve this qualification.

11. Throughout equipment training, the Trainee must maintain a passing grade of 60% on all written or practical Performance Checks, (PCs) and/or Enabling Checks (ECs) and/or Maintenance Procedural Training (MPT) scenarios. The passing grade for an enabling check is also 60%.

12. Enabling Checks (ECs). ECs are formative tests that help the Trainee and the instructors recognize progress or gaps in learning, so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. They also reinforce learning, so that it will be better retained for longer periods. Formative tests are designed to give feedback to both the instructor(s) and Trainees. Each EO must have at least one EC. Marks for ECs are not calculated in the Trainee's final mark.

13. Performance Checks (PCs). PCs are summative tests are used to determine if the Trainee has met the standard under the conditions (where possible) of the PO and hence can do the job. Summative tests can take the form of PCs or Critical ECs (reserved for safety related content). Supplemental PCs are administered upon failure of a PC or critical EC and must be in the same format and have the same standard as the original PC.

GRADING

There are two distinct marking methods for the various Performance Objectives. Each PC/EC shall be conducted in accordance with Annex K. If the candidate successfully completes the PC and the overall weighted mark for the PO is less than 60%, the final grade for the PO will be 60%. Depending upon the equipment availability and course duration, the following methods shall be applied:

(1) PC(s) = no less than 75%;

(2) EC(s), Projects and worksheets = no more than 25%;

(3) If no worksheet or EC is provided the PC will be worth 100%.

14. Instructors shall complete the Table "W Eng Sonar Course Mark" as demonstrated in the example below. Based upon the PO methods found in Para 13. The following Final marks were determined:

a. PO XXX Final mark of 85%. The following mark was determined(No EC was given);

PC Fault 85%
PO Mark 85 x 100% = 85%

b. PO XXX Final mark 84.35%. The following mark was determined;

Work sheets 91%
PC Fault 84%
Work sheets 91 x 5% = 4.55%
PC Fault 84 x 95%= 79.80%
_____
PO Mark = 84.35%

c. PO XXX Final mark 80.54%. Using Method Two, the following mark was determined;

EC Practical 80%
EC Practical 85%
Project 90%
Work sheets 89%
PC Fault 79%
Total EC Practical (80 + 85)/2 x 15%= 12.34%
Project 90 x 5 = 4.5%
Work sheet 89 x 5%= 4.45%
PC Fault 79 x 75% = 59.25%
_____
PO Mark = 80.54%

15. Each PO final mark shall be weighted according to the number of training days to determine overall mark. As per the example in the table below:

PO#	Training Days (TD)	Total Training Days (TTD)	PO Final Mark	Weighted Mark (PO Final Mark x TD) / TTD	
xxx	7	10	85	(85x7)/10	59.5
yyy	2		76	(76x2)/10	15.2
zzz	1		100	(100x1)/10	10

Overall Mark					84.7%

UNSATISFACTORY PROGRESS

16. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

17. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

18. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

19. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

20. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

21. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

22. Formal Course: A Canadian Forces Course Report, Form CF 377 and/or CF289 Certificate of Military Achievement, shall be prepared and distributed in accordance with DAOD 5031-9. The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

RESPONSIBILITY FOR TESTING

23. Standards provide quality control in designing, administering, and scoring all Performance Checks and Critical Enabling Checks. Regular Enabling Checks are the responsibility of the Instructional staff.

PROGRESS MONITORING

24. Trainee progress monitoring is done by instructors, instructor supervisors and if required, Standards and/or Quality Assurance. Monitoring, including unsatisfactory progress, is done IAW CFITES Vol 6 - Conduct of Instruction Programmes.

25. Course Senior Instructor shall:

a. perform Monthly Review Progress Reports and conduct interviews with each Trainee and;

b. ensure instructor(s) maintain up to date Trainee files containing results of the:

(1) PCs;

(2) ECs;

(3) projects;

(4) work/job sheets;

(5) home work; and

(6) progress Reports.

26. Performance Checks and Enabling Checks are based upon the POs and EOs as stated in Chapter 2 and 4 and are evaluated IAW Annex I.

27. To the greatest extent possible, PCs, and ECs shall be administered to the Trainee individually through the use of practical tests. Unavailability of equipment/MPT, or the unique nature of the subject matter, shall require the use of other forms of testing, for example, written tests.

28. A Trainee is allowed one (1) supplemental per EC, with a minimum of one (1) training day between the EC and the supplemental. The recheck shall be conducted under the supervision of the Senior Equipment Instructor. A pass on a supplemental will be assigned a grade of 60%. Regardless of the results of the recheck the marks will be retained on student file for use in an “unsatisfactory student progress” if needed and will be used in a TRB.

29. A Trainee is allowed one (1) supplemental per PC, with a minimum of one (1) training day between the PC and the supplemental. The supplemental shall be conducted under the supervision of the Senior Equipment Instructor. No supplemental assessment shall interfere with subsequent training and should be scheduled based on instructor/student availability. A pass on a supplemental will be assigned a grade of 60%. A failure of the supplemental constitutes “Unsatisfactory Training Progress”.

30. When possible, there shall be four different faulted areas in one practical PC/EC. These four faulted areas should be as follows:

a. Power distribution or equivalent (ie: fuse);

b. Hydraulic/mechanical (if applicable);

c. Initialization (ie: switches); and

d. Lowest replaceable unit (LRU).

31. Initialization ECs shall consist of switch(s) and/or control fault(s). The Trainee is required to fully initialize the system and report any equipment deficiencies to the assessor utilizing proper reporting procedures. LRUs may be used to indicate equipment deficiencies; however, the Trainee is only required to identify the deficiency.

CHAPTER 4 LESSON SPECIFICATIONS

PO 001 – Maintain Navigation Data Distribution System

EO 001.01 Operate a Navigation Data Distribution System

1. Performance. Operate a Navigation Data Distribution System (NDDS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to NDDS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or NDDS trainer; and

      (3) Day or night.

3. Standard. The trainee shall operate an NDDS IAW refs, by:

   a. Identifying the major components to the block diagram level, to include:

      (1) Data servers;

      (2) CDU; and

(3) Ancillary equipment.

b. Describing the functions of the major components;

c. Describing the redundancy features of the NDDS;

d. Identifying the power requirements of the system;

e. Initializing the NDDS, to include:

(1) Flashing up the system; and

(2) Configuring the system, to include:

(a) Selecting sensor sources; and

(b) Performing a functional check.

4. Time Required.

a. EO = 17 x 50 min periods.

5. Instructional Method.

a. Lecture;

b. Demonstration; and

c. Practice.

6. References.

a. A01 – A07; and

b. B01 – B02.

7. Training Aids.

a. NDDS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 001.01 will be tested with PC 001 IAW Annex H

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Identify NDDS components to include: a. Main & Aux NDDS description b. Navigational Feeds Ancillary equipment such as RICS, Servers, Converters, Panels	L D	1 0.5	
TP2	1. Initialize the NDDS to include: a. Flash-up procedure	L D	1 0.5	
TP3	1. Configure NDDS to meet Operational Requirements to include: a. Modes of operation	L D	1 0.5	
TP4	1. Perform NDDS function checks to include: a. NDDS Functional Test b. NDDS Alarm Panel Test c. Interface Port tests d. GPS and CMS Communication e. NDDS Server Test f. Main/Aux LAN Fail over Test	L D P	4 1 4	
TP5	1. Operate Navigation Data Distribution System to include: a. Verify operations	L D	1 0.5	

	b. Simulated outputs	P	2	

PO 001 – Maintain Navigation Data Distribution System

EO 001.02 Repair a Navigation Data Distribution System Fault

1. Performance. Repair a Navigation Data Distribution System (NDDS) Fault.

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to NDDS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or NDDS trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair an NDDS fault IAW refs, by:

   a. Operating the NDDS via the CDU, to include:

      (1) Flashing-up;

      (2) Verifying operational readiness;

      (3) Reporting status to supervisor; and

      (4) Powering down.

b. Identifying faults, to include:

(1) Checking assembly LEDs;

(2) Interpreting on-line BIT and fault codes;

(3) Viewing fault codes with the CDU;

(4) Conducting a CDU controlled off-line BIT; and

(5) Replacing/repairing the LRU.

c. Verifying repair to include:

(1) Initializing the NDDS to an operational state;

(2) Confirming correct operation of the NDDS; and

(3) Configuring a NDDS to meet an operation requirement.

d. Completing technical administration.

4. Time Required.

a. EO = 25.5 x 50 min periods; and

b. PC = 16 x 50 min periods.

5. Instructional Method.

a. Lecture;

b. Demonstration;

c. Practice; and

d. PC

6. References.

a. A01 – A07; and

b. B01 – B02.

7. Training Aids.

a. NDDS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 001.02 will be tested with PC 001 IAW Annex H; and

b. PC 001 and debrief to be conducted IAW Annex H with a 90 min time.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned;

c. Assignments are to be completed during non-instructional time; and

d. Two period for fault reset and pre/post brief.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Diagnose a NDDS to LRU by: a. use a CDU unit; b. follow a fault isolation table; and c. Offline/Online Bit Tests.	L D P	0.5 1 9.5	
TP2	1. Remove faulty component / install new component by: a. replacing a fuse b. replacing compact flash memory card c. remove/replace circuit cards to include. (1) GPS; (2) NTDS; and (3) PTFD. d. replacing a Battery	L	1	
TP3	1. Perform post repair test/ alignment on NDDS to include: a. restoring software b. configuration data c. jumper configurations d. IP addressing	L	1	
TP4	1. Perform NDDS Functional Checks to include: a. NDDS functional test; b. NDDS alarm panel test; c. interface port tests; d. GPS and CMS	L D P	4 1 4	

	Communication; e. NDDS server test; and f. Main/Aux LAN fail over test			
TP5	1. Operate NDDS to include: a. verify operations; and b. simulated outputs	L D P	1 0.5 2	

PO 002 – Maintain Inertial Navigation System

EO 002.01 Describe the Inertial Navigation System

1. Performance. Describe the Inertial Navigation System (INS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to INS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or INS trainer; and

      (3) Day or night.

3. Standard. The trainee shall describe the INS IAW refs, by:

   a. System operation;

   b. System configurations;

   c. System functions;

   d. Physical layout of equipment; and

e. Indicators and controls.

4. Time Required.

a. EO = 14.5 x 50 min periods.

5. Instructional Method.

a. Lecture;

b. Demonstration; and

c. Practice;

6. References.

a. A01 – A07; and

b. C01.

7. Training Aids.

a. INS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 002.01 will be tested with PC 002 IAW Annex H.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Describe the INS system operation to include: a. modes of operation b. capabilities and operations	L	1	
TP2	1. Describe system configuration to include: a. system block diagram; b. general signal flow; and c. system power requirements	L	2	
TP3	1. Describe system functions to include: a. Inertial Measurement Unit (IMU) to include: (1) Ring laser gyro theory; (2) Indexing; (3) Accelerometer; and (4) LIM voltages. b. IMU support electronics; c. Navigation, I/O processors, and interface electronics; d. Power supplies and battery backup; and e. Keypad and display	L	1	
TP4	1. Describe the physical layout of equipment to include; a. ship's physical arrangement;	D	1	

	and b. cabinet layout			
TP5	2. Indicators and controls to include: a. Familiarize with publications and gyro logs; b. Familiarization with location of controls and indicators; c. Describe built-in test equipment; and d. Describe test mode.	L D	0.5 2	

PO 002 – Maintain Inertial Navigation System

EO 002.02 Initialize the Inertial Navigation System

1. Performance. Initialize the Inertial Navigation System (INS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to INS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or INS trainer; and

      (3) Day or night.

3. Standard. The trainee shall initialize the INS IAW refs, by:

   a. Applying system power;

   b. Carrying out equipment functional checks;

   c. Verifying operational readiness;

   d. Reporting equipment status to supervisor; and

   e. Powering down.

4. Time Required.

   a. EO = 5 x 50 min periods.

5. Instructional Method.

   a. Demonstration; and

   b. Practice;

6. References.

   a. A01 – A07; and

   b. C01.

7. Training Aids.

   a. INS and/or MPT;

   b. Audio Visual aids; and

   c. Handouts.

8. Test Details.

   a. EO 002.02 will be tested as part of PC 002 IAW Annex H.

9. Remarks.

   a. MPT may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

   a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Initialize the INS to include: a. perform flash-up; b. verify that all controls and switches are in the correct position; and 2. Carry out equipment functional checks to include: a. verify inputs and outputs; b. verify alignment sequence 3. Verify operational readiness; 4. Report equipment status to supervisor; and 5. Power down.	D P	1 4	

PO 002 – Maintain Inertial Navigation System

EO 002.03 Repair the Inertial Navigation System

1. Performance. Repair the Inertial Navigation System (INS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to INS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or INS trainer; and

      (3) Day or night.

3. Standard. The trainee shall initialize the INS IAW refs, by:

   a. Performing flash-up;

   b. Localizing fault to LRU;

   c. Repairing and calibrating system; and

   d. Verifying operational readiness.

4. Time Required.

   a. EO = 16 x 50 min periods; and

   b. PC = 8 x 50 min periods.

5. Instructional Method.

   a. Practice.

6. References.

   a. A01 – A07; and

   b. C01.

7. Training Aids.

   a. INS and/or MPT;

   b. Audio Visual aids; and

   c. Handouts.

8. Test Details.

   a. EO 002.03 will be tested as part of PC 002 IAW Annex H.

9. Remarks.

   a. MPT may be utilized;

   b. Projects and exercises may be assigned;

   c. Assignments are to be completed during non-instructional time; and

   d. PC 002 and debrief will consist of 1 x 60 min period.

10. Teaching Points.

   a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	Maintain the INS 1. Initialize INS to include: a. localizing fault to LRU b. monitor fault indicators; and c. perform and interpret BITE test	P	16	
TP2	1. Perform troubleshooting, to include: a. Measure test points as necessary; and b. identify faulted LRU			
TP3	1. Repairing and aligning system a. replace faulted LRU b. perform alignment as necessary; c. monitor fault codes			
TP4	1. Verify operational readiness.			

PO 003 – Maintain Shipboard Integrated Navigation and Display System

EO 003.01 Describe the Shipboard Integrated Navigation and Display System

1. Performance. Describe the Shipboard Integrated Navigation and Display System (SHINNADS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SHINNADS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SHINNADS trainer; and

      (3) Day or night.

3. Standard. The trainee shall describe the SHINNADS IAW refs, by:

   a. System operation;

   b. System configurations;

   c. System functions;

   d. Physical layout of equipment; and

e. Software configuration.

4. Time Required.

a. EO = 13 x 50 min periods.

5. Instructional Method.

a. Lecture;

b. Demonstration; and

c. Practice;

6. References.

a. A01 – A07; and

b. D01.

7. Training Aids.

a. A SHINNADS console and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 003.01 will be tested with PC 003 IAW Annex H.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Describe the SHINNADS system operation to include; a. system operation; b. modes of operation; and c. capabilities and limitations	L	1	
TP2	1. Describe the SHINNADS System configuration to include; a. system block diagram; and b. general signal flow	L	2	
TP3	1. Describe the system's components to include: a. RTMON, RTPLAN and RTBACKUP; b. Starlink DGPS; c. Peripheral Back-up Switch; d. Sensor Interface Unit; e. KVM Switch; f. Operator Control Panel; g. power distribution and UPS; h. Radar Image Overlay; i. Serial Expansion Board; j. Ethernet Hub; k. video and display equipment; and l. fault isolation and repair	L	4	

	procedures			
TP4	1. Describe the physical layout of equipment to include: a. ship physical arrangement b. cabinet layout c. antenna location.	D	1	
TP5	1. Describe indicators and controls to include; a. Publication layout familiarization; and b. Controls and indicators location familiarization.	D	1	
TP6	1. Describe Software Configuration to include; a. Setup menu layout: b. Serial Comm port setup c. Radar Image Overlay setup d. Sensor setup; and e. Software Re-image Procedure	L D P	1 1 2	

PO 003 – Maintain Shipboard Integrated Navigation and Display System

EO 003.02 Initialize the Shipboard Integrated Navigation and Display System

1. Performance. Initialize the SHINNADS

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SHINNADS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SHINNADS trainer; and

      (3) Day or night.

3. Standard. The trainee shall initialize the SHINNADS IAW refs, by:

   a. Performing flash-up;

   b. Carrying out equipment functional checks;

   c. Verifying operational readiness;

   d. Reporting equipment status to supervisor; and

   e. Powering down.

4. Time Required.

   a. EO = 3 x 50 min periods.

5. Instructional Method.

   a. Demonstration; and

   b. Practice;

6. References.

   a. A01 – A07; and

   b. D01.

7. Training Aids.

   a. A SHINNADS console and/or MPT;

   b. Audio Visual aids; and

   c. Handouts.

8. Test Details.

   a. EO 003.02 will be tested with PC 003 IAW Annex H.

9. Remarks.

   a. MPT may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

   a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	Initialize the SHINNADS 1. Perform flash-up to include: a. Confirm all the controls and switches are in the correct orientation; and b. Following the proper initialization sequence. 2. Carry out equipment functional checks; and 3. Verify operational readiness.	D P	1 2	

PO 003 – Maintain Shipboard Integrated Navigation and Display System

EO 003.03 Repair the Shipboard Integrated Navigation and Display System

1. Performance. Repair the SHINNADS

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SHINNADS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SHINNADS trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair the SHINNADS IAW refs, by:

   a. Initializing the SHINNADS;

   b. Localizing fault to LRU;

   c. Repairing and aligning system; and

   d. Verifying operational readiness.

4. Time Required.

   a. EO = 16 x 50 min periods; and

   b. PC = 8 x 50 min periods.

5. Instructional Method.

   a. Practice; and

   b. PC;

6. References.

   a. A01 – A07; and

   b. D01.

7. Training Aids.

   a. A SHINNADS with system interfaes and/or MPT;

   b. Audio Visual aids; and

   c. Handouts.

8. Test Details.

   a. EO 003.03 will be tested with PC 003 IAW Annex H; and

   b. PC 003 and debrief will 1 x 50 min period time limit. (Correctie measures start at minute 45).

9. Remarks.

   a. MPT may be utilized;

   b. A signal flow project may be assigned;

   c. Assignments are to be completed during non-instructional time;

   d. Practice faultfinding on equipment will always be done in teams of 2.

10. Teaching Points.

   a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	Maintain the SHINNADS 1. Initialize the SHINNADS 2. Localizing fault to LRU to include; a. Monitor fault indicators; b. Utilize trouble-shooting flow charts; and c. Identify faulted LRU. 3. Repairing and aligning system to include; a. Replace faulted LRU; b. Monitor Fault Indicators; and c. Verify operational readiness.	P	16	

PO 004 – Maintain the Sonobuoy Processing System

EO 004.01 Describe the Sonobuoy Processing System

1. Performance. Describe the Sonobuoy Processing System (SPS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SPS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SPS trainer; and

      (3) Day or night.

3. Standard. The trainee shall describe the SPS IAW refs, by:

   a. System operation;

   b. System configurations;

   c. System functions;

   d. Physical layout of equipment; and

e. Indicators and controls.

4. Time Required.

a. EO = 6 x 50 min periods.

5. Instructional Method.

a. Lecture; and

b. Demonstration.

6. References.

a. A01 – A07; and

b. E01 – E05.

7. Training Aids.

a. SPS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 004.01 will be tested with PC 004 IAW Annex H.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Describe the SPS operation to include: a. System operation; b. Modes of operation; and c. Capabilities and limitations.	L	1	
TP2	1. Describe system configuration to include: a. System block diagram; and b. General signal flow.	L	1	
TP3	1. Describe system functions to include: a. The function of Receiver; b. The function of the Processor; and c. The function of the Display system.	L	2	
TP4	1. Describe the physical layout of equipment to include: a. Ship physical arrangement; b. Cabinet layout; and c. Antenna location.	D	1	

TP5	1. Indicators and controls to include: a. Familiarize with publication layout; b. Familiarize with location of controls and indicators; c. Describe built in test equipment; and d. Familiarize with accessing SPS via the Combat Management System (CMS).	L	1	

PO 004 – Maintain the Sonobuoy Processing System

EO 004.02 Initialize the Sonobuoy Processing System

1. Performance. Initialize the Sonobuoy Processing System (SPS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SPS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SPS trainer; and

      (3) Day or night.

3. Standard. The trainee shall initialize the SPS IAW refs, by:

   a. Powering up;

   b. Carrying out equipment functional checks;

   c. Verifying operational readiness;

   d. Reporting equipment status to supervisor; and

e. Powering down.

4. Time Required.

a. EO = 3 x 50 min periods.

5. Instructional Method.

a. Demonstration; and

b. Practice;

6. References.

a. A01 – A07; and

b. E01 – E05.

7. Training Aids.

a. SPS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 004.02 will be tested with PC 004 IAW Annex H.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Initialize the SPS: a. Power up to include; (1) verify that all controls and switches are in the correct position; and (2) perform Power up b. Carry out equipment functional checks; c. perform and interpret BITE tests; d. Verify operational readiness; e. Report equipment status to supervisor; f. Perform Power down.	D P	1 2	

PO 004 – Maintain the Sonobuoy Processing System

EO 004.03 Repair the Shipboard Integrated Navigation and Display System

1. Performance. Repair the Sonobuoy Processing System (SPS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SPS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SPS trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair the SPS IAW refs, by:

   a. Initializing the SPS;

   b. Localizing fault to LRU;

   c. Repairing and aligning system; and

   d. Verifying operational readiness.

4. Time Required.

   a. EO = 16 x 50 min periods; and

   b. PC = 8 x 50 min periods.

5. Instructional Method.

   a. Practice; and

   b. PC;

6. References.

   a. A01 – A07; and

   b. E01 – E05.

7. Training Aids.

   a. SPS and/or MPT;

   b. Audio Visual aids;

   c. Handouts; and

   d. References.

8. Test Details.

   a. EO 004.03 will be tested with PC 004 IAW Annex H; and

   b. PC 004 and debrief to be conducted with a 1 X 60 min period.

9. Remarks.

   a. MPT may be utilized;

   b. Projects and exercises may be assigned;

   c. Assignments are to be completed during non-instructional time;

   d. Practice faultfinding on equipment will always be done under supervision.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Initialize the SPS	P	16	
TP2	1. Localizing fault to LRU by: a. monitor fault indicators b. perform and interpret BITE c. utilize trouble-shooting flow charts d. identify faulted LRU			
TP3	1. Repairing and aligning system, by; a. lace faulted LRU; b. perform and interpret BITE			
TP4	1. Verify operational readiness by performing a. Loopback test.			

PO 005 – Maintain Canadian Towed Array Sonar System

EO 005.01 Describe Canadian Towed Array Sonar System

1. Performance. Describe the Canadian Towed Array Sonar System (CANTASS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to CANTASS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or CANTASS trainer; and

      (3) Day or night.

3. Standard. The trainee shall describe the CANTASS IAW refs, by:

   a. System operation;

   b. System configurations;

   c. System functions;

d. Physical layout of equipment; and

e. Indicators and controls.

4. Time Required.

a. EO = 15 x 50 min periods.

5. Instructional Method.

a. Lecture; and

b. Demonstration.

6. References.

a. A01 – A07; and

b. F01 – F17.

7. Training Aids.

a. CANTASS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 005.01 will be tested with PC 005 IAW Annex H.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Describe system operation to include: a. system operation; b. modes of operation; and c. capabilities and limitations.	L	1	
TP2	1. Describe system configuration to include: a. system block diagram; and b. general signal flow.	L	1	
TP3	1. Describe system functions to include: a. the function of the Array/Handling and Stowage Group; b. the function of the Array receiver; c. the function of the Shipboard Electronic Sub System (SESS); d. the Display Control Group; and e. the Maintenance console.	L	9	
TP4	1. Describe the physical layout of equipment to include: a. ship physical arrangement; and b. cabinet layout.	D	2	

TP5	1. Indicators and Controls to include: a. familiarize with publication layout; and b. Familiarize with location of controls and indicators.	D	2	

PO 005 – Maintain Canadian Towed Array Sonar System

EO 005.02 Initialize the Canadian Towed Array Sonar System

1. Performance. Initialize the Canadian Towed Array Sonar System (CANTASS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to CANTASS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or CANTASS trainer; and

      (3) Day or night.

3. Standard. The trainee shall initialize the CANTASS IAW refs, by:

   a. Performing flash-up;

   b. Loading and verifying software;

   c. Verifying equipment functional checks;

   d. Verifying system readiness;

e. Reporting equipment status to supervisor; and

f. Powering down in the correct sequence.

4. Time Required.

a. EO = 6 x 50 min periods.

5. Instructional Method.

a. Demonstration; and

b. Practice;

6. References.

a. A01 – A07; and

b. F01 – F17.

7. Training Aids.

a. CANTASS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 005.02 will be tested with PC 005 IAW Annex H.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Perform flash-up, to include: a. verify that all controls and switches are in the correct position; and b. initialize components in the correct sequence.	D P	2 4	
TP2	1. Load and verify software: a. load operational software; and b. verify successful load.			
TP3	1. Perform equipment functional checks by verifying monitor health.			
TP4	1. Verify system readiness at the DTS.			
TP5	1. Report equipment status to supervisor.			
TP6	1. Perform Power down in the correct sequence.			

PO 005 – Maintain Canadian Towed Array Sonar System

EO 005.03 Repair the Canadian Towed Array Sonar System

1. Performance. Repair the Canadian Towed Array Sonar System (CANTASS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to CANTASS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or CANTASS trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair the CANTASS IAW refs, by:

   a. Describing the Array/H&SG to include:

      (1) Power distribution;

      (2) Hydraulic system;

      (3) CANTASS Control Console;

(4) CANTASS Levelwind;

(5) Handling system; and

(6) Fault Isolation Test Set (FITS).

b. Repairing the above sub-systems, to include:

(1) Initializing Array/H&SG;

(2) Localizing fault to LRU;

(3) Reparing and aligning system; and

(4) Verifying operational readiness.

4. Time Required.

a. EO = 74 x 50 min periods; and

b. PC = 17 x 50 min periods.

5. Instructional Method.

a. Lecture;

b. Demonstration; and

c. Practice.

6. References.

a. A01 – A07; and

b. F01 – F17.

7. Training Aids.

a. CANTASS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 005.03 will be tested with PC 005 IAW Annex H; and

b. PC 005 and debrief to be conducted with a 90 min time limit.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned;

c. Assignments are to be completed during non-instructional time;

d. Practice faultfinding on equipment will always be done under supervision; and

e. Two periods added for PC pre-brief and fault reset.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Describe power distribution to include: a. 440 VAC; b. 115 VAC; and c. 24 VDC. 2. Describe hydraulic system to include: a. Pressure regulation; b. Temperature regulation; c. Winch operation; d. Pawl operation; and e. Brake operation. 3. Describe the CANTASS Control Console to include: a. Switch Gear Junction Box; b. Motor Controller Junction Box; c. Upper Control Panel; d. Lower Control Panel; e. Performance monitoring/Fault Location Panel; and f. Card Cage. 4. Describe the CANTASS Levelwind to include: a. Carriage Support Frame;	L D P	12 6 7	

	b. Encoder Assembly; c. Auxiliary Junction Box; d. Carriage Assembly; e. Bearing Track; and f. Main Junction Box. 5. Describe the CANTASS Handling System to include: a. Torque Motor; b. Speed Reducer; and c. Junction box.			
TP2	1. Maintain the Array/H&SG by: a. Initializing the Array/H&SG; b. Localizing faults to the LRU by: (1) Monitor fault indicators; and (2) Utilize troubleshooting flowcharts c. Repairing the system; and d. Verifying operational readiness.	P	49	

EO 005.04 Repair the Array Receiver

1. Performance. Repair the Array Receiver (AR).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to CANTASS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or CANTASS trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair the AR IAW refs, by:

   a. Describing the AR to include:

      (1) Power distribution;

      (2) Array Receiver Controller; and

      (3) Array Power Supply.

   b. Repairing the above sub-systems, to include:

(1) Initializing Array Receiver;

(2) Localizing fault to LRU;

(3) Repairing and aligning system; and

(4) Verifying operational readiness.

4. Time Required.

a. EO = 6 x 50 min periods.

5. Instructional Method.

a. Lecture;

b. Demonstration; and

c. Practice.

6. References.

a. A01 – A07; and

b. F01 – F17.

7. Training Aids.

a. AR with interfaces and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 005.04 will be tested with PC 005 IAW Annex H.

9. Remarks.

a. MPT may be utilized;

b. A signal flow project may be assigned;

c. Assignments are to be completed during non-instructional time; and

d. Practice faultfinding on equipment will always be done under supervision.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Describe the Array Receiver to include: a. power distribution; b. Array Receiver Controller; and c. Array Power Supply.	L D P	0.5 0.5 0.5	
TP2	1. Maintain the Array Receiver by: a. initializing the Array Receiver; b. run and interpret diagnostics; c. localizing fault to LRU to include: (1) monitor fault indicators; and (2) utilize trouble- shooting flow charts. d. repairing the system; and e. Verifying operational readiness.	P	4	

PO 005 – Maintain Canadian Towed Array Sonar System

EO 005.05 Repair the Shipboard Electronic Sub-System

1. Performance. Repair the Shipboard Electronic Sub-System (SESS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SESS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SESS trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair the SESS IAW refs, by:

   a. Describing the SESS to include:

      (1) Power distribution;

      (2) UPS;

      (3) DMD;

(4) HSSP;

(5) DSR/CCR;

(6) HDDR;

(7) DCG; and

(8) Diagnostic testing.

b. Repairing the above sub-systems, to include:

(1) Initializing SESS;

(2) Localizing fault to LRU;

(3) Repairing and calibration system; and

(4) Verifying operational readiness.

4. Time Required.

a. EO = 86 x 50 min periods.

b. EC = 10 x 50 min periods' and

c. PC = 19 x 50 min periods.

5. Instructional Method.

a. Lecture; and

b. Demonstration.

6. References.

a. A01 – A07; and

b. F01 – F17.

7. Training Aids.

a. CANTASS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 005.05 will be tested with PC 005 IAW Annex H;

b. EC 005.05 and debrief to be conducted with 60 min period time limit; and

c. PC 005 and debrief to be conducted with 60 min period time limit.

9. Remarks.

a. Eight period shall be allocated for ECs and sixteen periods shall be allocated for PCs;

b. MPT may be utilized;

c. Projects and exercises may be assigned;

d. Assignments are to be completed during non-instructional time;

e. EC 005.05 shall consist of 6 x 50 min periods allocated for SESS practice faults;

f. 4 x 50 min periods shall be allocated for overall CANTASS practice faults prior to commencing PC 005;

g. Two periods added for EC brief and fault reset; and

h. Three periods added for PC brief and fault reset.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Describe the SESS power distribution; 2. Describe the following: a. DMD; b. HSSP; c. DSR/CCR;	L D	32.5 4.5	

	d. HDDR; e. DCG; and f. Diagnostic testing.	P	8.5	
TP2	1. Maintain the SESS by: a. Initializing the SESS; b. Run and interpret diagnostics; c. Localizing faults to the LRU using available troubleshooting resources; d. Repairing the system; and e. Verifying operational readiness.	P	40	

PO 006 – Maintain Hull Mounted SONAR

EO 006.01 Describe the Hull Mounted SONAR

1. Performance. Describe the Hull Mounted Sonar (HMS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to HMS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or HMS trainer; and

      (3) Day or night.

3. Standard. The trainee shall describe the HMS IAW refs, by:

   a. System operation;

   b. System configurations;

   c. System functions;

   d. Physical layout of equipment;

e. Indicators and controls. And

f. SONAR trials, to include:

(1) Sonar Performance Figure (SPF);

(2) Power output checks; and

(3) Impedance checks.

4. Time Required.

a. EO = 15 x 50 min periods.

5. Instructional Method.

a. Lecture; and

b. Demonstration.

6. References.

a. A01 – A07; and

b. G01 – G06.

7. Training Aids.

a. HMS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 006.01 will be tested with PC 006 IAW Annex H.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Describe HMS System operation to include: a. System operation; b. Modes of operation; and c. Capabilities and limitations.	L	2	
TP2	1. Describe system configuration to include: a. System block diagram; and b. General signal flow	L	2	
TP3	1. Describe system functions to include: a. The function of the transducer, T/R switch, and dummy load; b. The function of the transmitter and power supply; c. The function of the SCC; d. The SP; e. The SOC; and f. The maintenance console.	L	9	
TP4	1. Describe physical layout of equipment to include: a. ship physical arrangement; and b. cabinet layout.	D	0.5	

TP5	1. Indicators and controls to include: a. familiarize with publication layout; and b. Familiarize with location of controls and indicators.	D	0.5	
TP6	1. SONAR trials	L	1	

PO 006 – Maintain Hull Mounted Sonar

EO 006.02 Initialize the Hull Mounted SONAR

1. Performance. Initialize the Hull Mounted Sonar (HMS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to HMS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or HMS trainer; and

      (3) Day or night.

3. Standard. The trainee shall initialize the HMS IAW refs, by:

   a. Powering up in the correct sequence;

   b. Loading and verifying software;

   c. Verifying equipment functional checks;

   d. Verifying system readiness;

e. Reporting equipment status to supervisor; and

f. Powering down in the correct sequence

4. Time Required.

a. EO = 3 x 50 min periods.

5. Instructional Method.

a. Demonstration; and

b. Practice;

6. References.

a. A01 – A07; and

b. G01 – G06.

7. Training Aids.

a. HMS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 006.02 will be tested with PC 006 IAW Annex H.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Initialize the HMS to include: a. Power Up in the correct sequence; b. Verify that all controls and switches are in the correct position; and c. Perform flash-up 2. Perform equipment functional checks by verifying monitor health; 3. Verify system readiness at the SOC or CMS; 4. Report system status to supervisor; and 5. Perform system shut down.	D P	1 2	

PO 006 – Maintain Hull Mounted SONAR

EO 006.03 Repair the Hull Mounted SONAR

1. Performance. Repair the Hull Mounted SONAR (HMS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to HMS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or HMS trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair the HMS IAW refs, by:

   a. Describing TX to include:

      (1) Power distribution;

      (2) TX power supply;

      (3) TX unit;

(4) T/R switching unit and dummy load; and

(5) Transducer.

b. Maintaining the above sub-systems to include:

(1) Initializing HMS;

(2) Localizing fault to LRU;

(3) Repairing and aligning system; and

(4) Verifying operational readiness.

4. Time Required.

a. EO = 12.5 x 50 min periods; and

b. PC = 1 x 50 min periods.

5. Instructional Method.

a. Lecture;

b. Demonstration; and

c. Practice.

6. References.

a. A01 – A07; and

b. G01 – G06.

7. Training Aids.

a. HMS and/or MPT;

b. Audio visual aids;

c. Handouts; and

d. References.

8. Test Details.

a. EO 006.03 will be tested as pat of EC 006.03 and PC 006 IAW Annex H; and

b. EC 006.03 and debrief to be conducted with 60 min time limit.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Describe TX power distribution. 2. Describe TX power supply. 3. Describe TX unit to include: a. Integrated test equipment; b. SONAR power inverters; c. Power supplies; d. T/R switching unit and dummy load; and e. Transducer.	L D	5.5 8	
TP2	1. Maintain the TX to include: a. Initializing the HMS; b. Run and interpret diagnostics; c. Localizing faults to the LRU using available troubleshooting resources; d. Repairing the system; and e. Verifying operational readiness.	P	6	

PO 006 – Maintain Hull Mounted SONAR

EO 006.04 Repair the SONAR Operating Console

1. Performance. Repair the SONAR Operating Console (SOC).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to HMS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or HMS trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair the SOC IAW refs, by:

   a. Describing the SOC to include:

      (1) Power distribution;

      (2) Television Monitor (TVM);

      (3) Control Panel Audio (CPA);

(4) Control Panel Remote (CPR);

(5) Power Junction Box (PJB);

(6) Control Unit Display (CUD); and

(7) High Density Digital Recorder (HDDR).

b. Repairing the above sub-systems to include:

(1) Initializing HMS;

(2) Localizing fault to LRU;

(3) Repairing and aligning system; and

(4) Verifying operational readiness.

4. Time Required.

a. EO = 6.5 x 50 min periods.

5. Instructional Method.

a. Lecture; and

b. Practice.

6. References.

a. A01 – A07; and

b. G01 – G06.

7. Training Aids.

a. HMS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 006.04 will be tested with PC 006 IAW Annex H.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned;

c. Assignments are to be completed during non-instructional time; and

d. Practice faultfinding on equipment will always be done under supervision.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Describe the SOC's power distribution. 2. Describe the following: a. TVM; b. CPA; c. CPR; d. PJB; e. CUD; and f. HDDR.	L D	1.5 1	
TP2	1. Maintain the SOC by: a. Initializing HMS; b. Run and interpret diagnostics; c. Localizing faults to the LRU using available troubleshooting resources;	P	4	

	d. Repairing the system; and e. Verifying operational readiness			

PO 006 – Maintain Hull Mounted SONAR

EO 006.05 Repair the Sonar Control and Processing Group

1. Performance. Repair the SONAR Control and Processing Group (CPG).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to HMS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or HS trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair the SONAR CPG IAW refs, by:

   a. Describing CPG to include:

      (1) Power distribution;

      (2) Sonar Control Cabinet (SCC);

      (3) Signal Processor (SP); and

(4) Maintenance Console (MC).

b. Repairing the above sub-systems, to include:

(1) Initializing HMS;

(2) Localizing fault to LRU;

(3) Repairing and aligning system; and

(4) Verifying operational readiness.

4. Time Required.

a. EO = 30.5 x 50 min periods.

b. EC = 1 x 50 min periods' and

c. PC = 1 x 50 min periods.

5. Instructional Method.

a. Lecture;

b. Demonstration; and

c. Practice.

6. References.

a. A01 – A07; and

b. G01 – G06.

7. Training Aids.

a. HMS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 006.05 will be tested with PC 006 IAW Annex H;

b. EC 006.05 and debrief to be conducted with 60 min period time limit; and

c. PC 006 and debrief to be conducted with 60 min period time limit.

9. Remarks.

a. MPT may be utilized;

b. A signal flow project may be assigned;

c. Assignments are to be completed during non-instructional time;

d. Practice faultfinding on equipment will always be done under supervision;

e. EO 006.05 shall include 2 x 60 min practice faults allotted for the CPG; and

f. PC 006 shall include 2 x 60 min overall HMS System practice faults prior to PC assessment.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	3. Describe the CPG to include: a. Power distribution; .	L	0.5	
	b. Describe the following: (1) SCC;	L	10	
	(2) SP; and	L	3	
	(3) MC.	L D P	4 1 4	

TP2	1. Maintain the CPG by: a. initializing the HMS; b. run and interpret diagnostics; c. localizing fault to LRU to include: (1) monitor fault indicators (2) utilize trouble-shooting flow charts d. repairing the system; and e. Verifying operational readiness.	P	8	

PO 007 – Maintain the Torpedo Decoy System

EO 007.01 Describe the Torpedo Decoy System

1. Performance. Describe the Torpedo Decoy System (TDS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to TDS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) HMC ship or TDS trainer; and

      (2) Day or night.

3. Standard. The trainee shall describe the TDS IAW refs, by:

   a. System operation;

   b. System configurations;

   c. System functions;

   d. Physical layout of equipment;

   e. Indicators and controls; and

f. Launch procedures.

4. Time Required.

a. EO = 15 x 50 min periods.

5. Instructional Method.

a. Lecture; and

b. Demonstration.

6. References.

a. A01 – A07; and

b. H01 – H03.

7. Training Aids.

a. TDS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 007.01 will be tested with PC 007 IAW Annex H.

9. Remarks.

a. Projects and exercises may be assigned; and

b. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. System Operation to include: a. Describe system operation; and b. Describe capabilities and limitations.	L	0.5	
TP2	1. System Configuration to include: a. Describe system block diagram; and b. Describe general signal flow.	L	0.5	
TP3	1. Describe the System Functions to include: a. Winch and slip ring assembly to include motor controller;	L D	2 1	
	b. Fiber Optic Tow cable (FOTC);	L	0.5	
	c. TB-14A Towed Body;	L	0.5	
	d. Electronic Console to include: (1) Signal generator drawer; (2) Output control drawer; (3) Transformer drawer; and	L D	4 2	

	(4) The Remote Control Unit (RCU).			
TP4	1. Describe physical layout of equipment to include; a. ship physical arrangement; and b. Cabinet layout.	D	1	
TP5	1. Indicators and controls to include: a. familiarize with publication layout; b. familiarize with location of controls and indicators; and c. describe built in test equipment.	L D	1.5 0.5	
TP6	1. Launch procedures	L	1	

PO 007 – Maintain the Torpedo Decoy System

EO 007.02 Initialize the Torpedo Decoy System

1. Performance. Initialize the Torpedo Decoy System.

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to TDS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or TDS trainer; and

      (3) Day or night.

3. Standard. The trainee shall initialize the TDS IAW refs, by:

   a. Powering up;

   b. Carrying out equipment functional checks;

   c. Verifying operational readiness;

   d. Reporting equipment status to supervisor; and

e. Powering down.

4. Time Required.

a. EO = 1 x 50 min periods.

5. Instructional Method.

a. Demonstration.

6. References.

a. A01 – A07; and

b. H01 – H03.

7. Training Aids.

a. TDS and/or MPT;

b. Audio Visual aids; and

c. Handouts.

8. Test Details.

a. EO 007.02 will be tested with PC 007 IAW Annex H.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Initialize the NIXIE to include: a. Power Up to include: (1) verify that all controls and switches are in the correct position; and (2) perform Power up. b. Carry out equipment functional checks by performing and interpreting BITE tests; c. Verify operational readiness by checking status indicators; d. Report equipment status to supervisor; and e. Perform Power down.	D	1	

PO 007 – Maintain the Torpedo Decoy System

EO 007.03 Repair the Torpedo Decoy System

1. Performance. Repair the Torpedo Decoy System.

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to TDS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or TDS trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair the TDS IAW refs, by:

   a. Initializing the TDS;

   b. Localizing fault to LRU;

   c. Repairing and aligning system; and

   d. Verifying operational readiness.

4. Time Required.

   a. EO = 12 x 50 min periods; and

   b. PC = 8 x 50 min periods.

5. Instructional Method.

   a. Practice.

6. References.

   a. A01 – A07; and

   b. H01 – H03.

7. Training Aids.

   a. TDS and/or MPT;

   b. Audio Visual aids; and

   c. Handouts.

8. Test Details.

   a. EO 007.03 will be tested with PC 007 IAW Annex H; and

   b. PC 007 and debrief to be conducted with 60 min time limit.

9. Remarks.

   a. MPT may be utilized;

   b. Projects and exercises may be assigned;

   c. Assignments are to be completed during non-instructional time;

   d. Practice faultfinding on equipment will always be done under supervision.

10. Teaching Points.

   a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Initialize the TDS	P	12	
TP2	1. Localize fault to LRU by; a. Monitor Fault indications; b. Run and interpret offline BITE; and c. Identify faulted LRU.			
TP3	1. Repairing and aligning system by replacing faulted LRU			
TP4	1. Verify operational readiness.			

PO 008 – Maintain the Shipboard Electro Optic Surveillance System

EO 008.01 Describe the Shipboard Electro Optic Surveillance System

1. Performance. Describe the Shipboard Electro Optic Surveillance System (SEOSS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SEOSS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SEOSS trainer; and

      (3) Day or night.

3. Standard. The trainee shall describe the SEOSS IAW refs, by:

   a. System operation;

   b. System configurations;

   c. Physical layout of equipment; and

   d. Indicators and controls.

4. Time Required.

   a. EO = 14 x 50 min periods.

5. Instructional Method.

   a. Lecture;

   b. Demonstration; and

   c. Practice.

6. References.

   a. A01 – A07; and

   b. I01 – I02.

7. Training Aids.

   a. SEOSS and/or MPT;

   b. Audio Visual aids; and

   c. Handouts.

8. Test Details.

   a. EO 008.01 will be tested with PC 008 IAW Annex H.

9. Remarks.

   a. MPT may be utilized;

   b. Projects and exercises may be assigned; and

   c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

   a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Describe system operation to include: a. System operation; b. Modes of operation; and c. Capabilities and limitations; and d. Publication layout familiarization.	L	1	
TP2	1. Describe system configurations to include: a. System block diagram; and b. General signal flow.	L	1	
TP3	1. Describe System Functions to include: a. Operator Stations main/remote; b. Hand controllers; c. Junction Box; d. Air Dryer Unit; e. Wash tank and pump; and f. Turret assembly.	L	3	
TP4	1. Physical layout of equipment a. Describe ship's physical layout (practice to include hand controllers)	L D P	1 1 2	

TP5	1. Indicators and controls to include a. Location of controls and indicators; and b. Describe built in test equipment.	L D P	2 1 2	
TP6	1. Laser Illuminator to include: a. Properties of light and lasers b. Classes of lasers c. Safety; and d. Activating and firing the laser.	L	3	

PO 008 – Maintain Shipboard Electro Optic Surveillance System

EO 008.02 Initialize the Shipboard Electro Optic Surveillance System

1. Performance. Initialize the Shipboard Electro Optic Surveillance System (SEOSS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SEOSS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SEOSS trainer; and

      (3) Day or night.

3. Standard. The trainee shall initialize the SEOSS IAW refs, by:

   a. Performing flash-up;

   b. Carrying out equipment functional checks;

   c. Verifying operational readiness;

   d. Reporting equipment status to supervisor; and

e. Powering down.

4. Time Required.

a. EO = 5 x 50 min periods.

5. Instructional Method.

a. Demonstration; and

b. Practice.

6. References.

a. A01 – A07; and

b. I01 – I02.

7. Training Aids.

a. SEOSS and/or MPT;

b. Audio Visual aids;

c. Handouts; and

d. References.

8. Test Details.

a. EO 008.02 will be tested with PC 008 IAW Annex H.

9. Remarks.

a. MPT may be utilized;

b. Projects and exercises may be assigned; and

c. Assignments are to be completed during non-instructional time.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Flash-up to include: a. Verifying that all controls and switches are in the correct position	D P	1 4	
TP2	1. Carry out equipment functional checks by performing BITs and interpreting results			
TP3	1. Verify operational readiness by checking status indicators			
TP4	1. Report equipment status to supervisor			
TP5	1. Perform Power down			

PO 008 – Maintain Shipboard Electro Optic Surveillance System

EO 008.03 Repair the Shipboard Electro Optic Surveillance System

1. Performance. Repair the Shipboard Electro Optic Surveillance System (SEOSS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SEOSS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SEOSS trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair the SEOSS IAW refs, by:

   a. Initializing the SEOSS;

   b. Localizing fault to LRU;

   c. Repairing and aligning system; and

   d. Verifying operational readiness.

4. Time Required.

   a. EO = 12 x 50 min periods; and

   b. PC = 10 x 50 min periods.

5. Instructional Method.

   a. Practice.

6. References.

   a. A01 – A07; and

   b. I01 – I02.

7. Training Aids.

   a. SEOSS and/or MPT;

   b. Audio Visual aids; and

   c. Handouts.

8. Test Details.

   a. EO 008.03 will be tested with PC 008 IAW Annex H; and

   b. PC 008 and debrief to be conducted with 60 min time limit.

9. Remarks.

   a. MPT may be utilized;

   b. Projects and exercises may be assigned;

   c. Assignments are to be completed during non-instructional time; and

   d. Two periods added for PC pre-brief and fault reset.

10. Teaching Points.

   a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Maintain SEOSS to include: a. initializing SEOSS; b. localizing fault to LRU by: (1) monitor fault indications; (2) running and interpreting offline BITE; and (3) identifying faulted LRU. c. repairing and aligning system by replacing faulted LRU; and d. verifying operational readiness	P	12	

PO 009 – Maintain Combat Management System 330

EO 009.01 Describe the Combat Management System 330

1. Performance. Describe the Combat Management System (CMS 330).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to CMS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or CMS trainer; and

      (3) Day or night.

3. Standard. The trainee shall describe the CMS IAW refs, by:

   a. System components;

   b. Capabilities and limitations;

   c. Theory of operation;

   d. Initial power distribution; and

e. Indicators and controls.

4. Time Required.

a. EO = 10 x 50 min periods.

5. Instructional Method.

a. Lecture.

6. References.

a. A01 – A07; and

b. J01 – J12.

7. Training Aids.

a. CMS and/or MPT;

b. Audio Visual aids;

c. Handouts; and

d. References.

8. Test Details.

a. EO 009.01 will be tested with PC 009 IAW Annex H.

9. Remarks.

a. For the purpose of this EO, the student is assumed to have a working knowledge of CMS operation, initialization, and maintenance. This training should be focused on the higher functions of the CMS.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Describe system components down to the LRU to include; a. Mission critical enclosures; b. Multi-function workstation; and, c. Local area network.	L	2	
TP2	1. Capabilities and limitations, including the consequences of loss, to include; a. Cabinet enclosures; b. Remote access server; c. Cabinet cooling system; and d. MCE cabinets.	L	2	
TP3	1. Theory of operation to include; a. layered system of sensors; b. information exchange systems(IES); c. tracking tools; and d. decision aids and weapons to include: (1) detect (2) locate; (3) track; (4) identify; (5) assess; and (6) engage threats.	L	2	
TP4	1. Power Distribution to include; a. breaker and switchboard connection; b. MCE power;	L	2	

	c. MFW power; and d. Subassembly voltage input.			
TP5	1. Indicators and controls to include; a. Common MCE Components; b. MCEs and their subassemblies. c. An understanding of the GUI to differentiate operator error versus faulty hardware/software component.	L	2	

PO 009 – Maintain Combat Management System 330

EO 009.02 Initialize the Combat Management System 330

1. Performance. Initialize the Combat Management System (CMS 330).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to CMS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or CMS trainer; and

      (3) Day or night.

3. Standard. The trainee shall initialize the CMS IAW refs, by:

   a. Powering up;

   b. Carrying out equipment functional checks;

   c. Loading CMS software;

   d. Verify indicator status;

e. Reporting equipment status to supervisor; and

f. Powering down.

4. Time Required.

a. EO = 10 x 50 min periods.

5. Instructional Method.

a. Lecture; and

b. Demonstration.

6. References.

a. A01 – A07; and

b. J01 – J12.

7. Training Aids.

a. CMS and/or MPT;

b. Audio Visual aids;

c. Handouts; and

d. References.

8. Test Details.

a. EO 009.02 will be tested with PC 009 IAW Annex H.

9. Remarks.

a. Practice flash-up and faultfinding will be conducted under supervision; and

b. EC/PC will be carried out individually.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Initialize CMS to include; a. Preliminary checks on associated equipment; b. Power up from cold boot; c. Power up from warm boot & software reload d. Checks during and after power on; e. Load and verify software; f. Conduct system functional checks; g. Verify operational readiness; h. Report System Status to supervisor; and i. Shutdown from operational load/unload	D P	2 8	

PO 010 – Maintain the Torpedo Handling System

EO 010.01 Operate the Torpedo Handling System

1. Performance. Operate the Torpedo Handling System (THS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to THS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or THS trainer; and

      (3) Day or night.

3. Standard. The trainee shall describe the THS IAW refs, by:

   a. Flash-up; and

   b. Configuring Torpedo Magazine.

4. Time Required.

   a. EO = 10 x 50 min periods; and

b. EC = 1 x 20 min period.

5. Instructional Method.

a. Lecture; and

b. Demonstration.

6. References.

a. A01 – A07; and

b. K01 – K05.

7. Training Aids.

a. STG approved Safety Brief;

b. Flash-up checklist;

c. Functional check checklist;

d. Pre-sail checklist;

e. OTTO fuel II SDS handout;

f. Unlabeled torpedo parts handout;

g. Ammunition colour coding handout;

h. Inspection upon receipt checklist;

i. Mock torpedo book; ad

j. HMC ship (as required).

8. Test Details.

a. EO 010.01 will be tested with PC 010 IAW Annex H.

9. Remarks.

a. If no trainer is available, instructor will need to coordinate with HMC ship for use of onboard system.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice

	Description	Method	Time	Reference
TP1	1. Define the purpose of the THS and its importance in response to an underwater threat.	L	1	
TP2	1. Identifying the major components of the THS to the block diagram level including the purpose of each component.	L	1	
TP3	1. Explaining the Controls and Indicators of the THS; 2. Identifying the safety requirements of the system to include: a. Fluid power; b. Mechanical; and c. PPE. 3. Identifying the rules and regulations involved with moving a torpedo to include: a. NAVORDS; b. NCPM 1001; and c. Ammunition Regs. 4. Performing Flash-up of a THS; 5. Performing Functional Checks on the THS; 6. Confirming weight certifications aren't expired on: a. Rails; b. Straps; c. Crane; and d. Loading Tray. 7. Performing pre-sail and 48-hour pre-sail checks;	L	8	

	8. Securing the system.			

PO 010 – Maintain the Torpedo Handling System

EO 010.02 Transport a Torpedo

1. Performance. Transport a Torpedo.

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to THS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or THS trainer; and

      (3) Day or night.

3. Standard. The trainee shall transport a torpedo IAW refs, by:

   a. Configuring torpedo magazine;

   b. Transferring a torpedo from the flight deck to the magazine;

   c. Stowing a torpedo in the torpedo magazine;

   d. Transferring a torpedo from the magazine to the flight deck; and

e. Completing required administration.

4. Time Required.

a. EO = 14 x 50 min periods; and

b. EC = 1 x 20 min period.

5. Instructional Method.

a. Lecture; and

b. Demonstration.

6. References.

a. A01 – A07; and

b. K01 – K05.

7. Training Aids.

a. STG approved Safety Brief;

b. Flash-up checklist;

c. Functional check checklist;

d. Pre-sail checklist;

e. OTTO fuel II SDS handout;

f. Unlabeled torpedo parts handout;

g. Ammunition colour coding handout;

h. Inspection upon receipt checklist;

i. Mock torpedo book; ad

j. HMC ship (as required).

8. Test Details.

a. EO 010.02 will be tested with PC 010 IAW Annex H.

9. Remarks.

a. If no trainer is available, instructor will need to coordinate with HMC ship for use of onboard system.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Configuring the torpedo magazine to include identifying and setting up peripheral equipment required for movement of torpedo.	D	1	
TP2	1. Deliver a safety brief; 2. Reviewing the inspection of a torpedo upon receipt; 3. Moving a torpedo from the flight deck to the securing chocks; 4. Moving a torpedo from the securing chocks to the flight deck; 5. Securing the evolution.	D	1	
TP3	1. Making an entry in the torpedo log; 2. Logging training time; 3. Listing potential administration to include: a. CF 140; b. Witness statements; c. CF98 entries; d. DND 663; and e. PEL entries.	L	0.5	

PO 010 – Maintain the Torpedo Handling System

EO 010.03 Repair the Torpedo Handling System

1. Performance. Perform Corrective Maintenance on the Torpedo Handling System (THS).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to THS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or THS trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair the THS IAW refs, by:

   a. Performing system diagnostics by:

      (1) Investigating System status;

      (2) Utilizing the troubleshooting guide; and

      (3) Identifying faulted component.

b. Repairing failed components in the system by:

(1) Producing a new Lowest Replaceable Unit (LRU);

(2) Replacing the LRU;

(3) Restoring operational capability; and

(4) Complete technical administration.

4. Time Required.

a. EO = 14 x 50 min periods; and

b. PC = 1 x 20 min period.

5. Instructional Method.

a. Lecture; and

b. Demonstration.

6. References.

a. A01 – A07; and

b. K01 – K05.

7. Training Aids.

a. STG approved Safety Brief;

b. Flash-up checklist;

c. Functional check checklist;

d. Pre-sail checklist;

e. OTTO fuel II SDS handout;

f. Unlabeled torpedo parts handout;

g. Ammunition colour coding handout;

h. Inspection upon receipt checklist;

i. Mock torpedo book; ad

j. HMC ship (as required).

8. Test Details.

a. EO 010.03 will be tested with EC 010.03.

9. Remarks.

a. If no trainer is available, instructor will need to coordinate with HMC ship for use of onboard system.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Confirming System status though: a. Conducting functional check; b. Operating the system; and c. Determining if the system is INOP/Degraded/Operation.	L	1	
TP2	1. Reporting the system status to STOW and/or CSEO using “SIOR”, “Situation, Impact, Options, Repair Plan”	L	1	
TP3	1. Utilizing the THS Fault Logic Diagrams (FLD) to help locate the faulted component.	L	1	
TP4	1. Using the Illustrated Parts Breakdown (IPB) to obtain replacement part information.	L	1	
TP5	1. Replacing LRU to include: a. Identify potential hazards; b. Identify the test/alignments/adjustments required for the replacement part; and c. Repair/replace faulted LRU.	L	1	
TP6	1. Conducting post repair testing to ensure system operability. 2. Reporting system status; 3. Completing technical administration.	L	1	

PO 010 – Maintain the Torpedo Handling System

EO 010.04 Direct Torpedo Evolutions

1. Performance. Direct Torpedo Evolutions.

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to THS or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or THS trainer; and

      (3) Day or night.

3. Standard. The trainee shall direct torpedo evolutions IAW refs, to include:

      (1) Conducting pre-operational checks;

      (2) Direct the load of a torpedo; and

      (3) Direct the unload of a torpedo.

4. Time Required.

a. EO = 14 x 50 min periods; and

b. PC = 1 x 20 min period.

5. Instructional Method.

a. Lecture; and

b. Demonstration.

6. References.

a. A01 – A07; and

b. K01 – K07.

7. Training Aids.

a. STG approved Safety Brief;

b. Flash-up checklist;

c. Functional check checklist;

d. Pre-sail checklist;

e. OTTO fuel II SDS handout;

f. Unlabeled torpedo parts handout;

g. Ammunition colour coding handout;

h. Inspection upon receipt checklist;

i. Mock torpedo book; ad

j. HMC ship (as required).

8. Test Details.

a. EO 010.04 will be tested with EC 010.03 and PC 010 IAW Annex H.

9. Remarks.

a. If no trainer is available, instructor will need to coordinate with HMC ship for use of onboard system.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Conduct pre-operational checks, to include: a. Ensure availability of power and required external systems; b. Inspect certification dates of required equipment; c. Ensure any required tools and references are available; and d. Obtain permission from command to operate Torpedo Weapon System.	D	1	
TP2	1. Conduct flash-up, to include: a. Apply power; b. Configure the system for desired operating environment; c. Configure external fire control systems; d. Ensure the system is in the desired safety configuration; and e. Secure the system at the end of operation;	D	1	
TP3	1. Perform functional checks, to include: a. Confirm expected operation of controls and indicators; b. Connect and reconfigure any required test devices; and c. Verify connectivity with external firing and control systems	D	8	
TP4	1. Load and unload a torpedo under direction to include: a. Understand the words of command; b. Explain the roles of the torpedo handling crew;	D	1	

	c. Explain required safety information; and d. Act as a member of the torpedo handling crew under supervision.			
TP5	1. Direct a torpedo handling crew to include: a. Ensure torpedo magazine is prepared; b. Ensure that all required tools and ancillary items are available; c. Conduct a safety brief; d. Assign crew roles; e. Load and unload a training torpedo; and f. Secure the magazine and torpedo weapon system.	P	16	

PO 011 – Maintain Surface Vessel Torpedo Tube

EO 011.01 Complete a Weapons Certification Test

1. Performance. Complete a Surface Vessel Torpedo Tube (SVTT) weapon certification test.

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SVTT or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SVTT trainer; and

      (3) Day or night.

3. Standard. The trainee shall describe the SVT IAW refs.

4. Time Required.

   a. EO = 14 x 50 min periods; and

   b. EC = 1 x 20 min period.

5. Instructional Method.

a. Lecture; and

b. Demonstration.

6. References.

a. A01 – A07; and

b. L01 – L03.

7. Training Aids.

a. STG Approved Safety Brief;

b. Flash-up checklist;

c. Functional check checklist;

8. Test Details.

a. EO 011.01 will be tested as a part of EC 011.01.

9. Remarks.

a. Nil.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Provide SVTT introduction lecture to include: a. Provide an overview of the system and its capabilities; b. Explain the role it has in the ASW suite and the impact on the ship if it is lost; c. Identify all the major components; d. Identify all the relevant publications; and e. Provide an overview of each task associated with operating SVTT.	L	1	
TP2	1. Identifying and defining each state of preparedness, to include: a. Standing To; b. Relaxed; c. Stood Down; and d. Secured. 2. Explaining Mechanically safe and electrically safe; 3. Explaining the state of the weapon system and crewing during the following: a. Firs degree of readiness; b. Second degree of readiness; and c. Third/Fourth degree of readiness; 4. Reviewing NAVORD Weapons Keys Policy WRT SVTT.	L	1	

TP3	1. Identifying the types of torpedos used by the RCN and their purpose, to include: a. WarShot; (1) Mk 46; (2) Mk 48; and (3) Mk 54. b. Exercise Torpedo; and c. HOTTORP.	L	1	
TP4	1. Describing torpedo safety to include: a. Safety for personnel such as: (1) Sharp propeller blades; (2) Wires; (3) Weight and (4) Explosive. b. Safety for the torpedo such as: (1) Do not climb/sit on; (2) Protecting the Transducer; (3) Avoiding scratching the torpedo body; and (4) Do not move by fins. 2. Identifying risks from Otto fuel: a. Describing Otto fuel; b. Reviewing SDS; c. Initial Hazmat response to Otto fuel leak/spill to include:	L	1	

	(1) Stopping the leak; (2) Space evacuation; and (3) Immediate ventilation. d. Using the Otto fuel detector. 3. Assessing hazardous situations to include: a. “A” appears in the indicator window of the exploder; b. Seawater battery arming lanyard removed; c. Motor start; d. Torpedo dropped in water; and e. Torpedo dropped on deck.			

PO 011 – Maintain Surface Vessel Torpedo Tube

EO 011.02 Operate a Surface Vessel Torpedo Tube

1. Performance. Operate a Surface Vessel Torpedo Tube (SVTT).

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SVTT or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SVTT trainer; and

      (3) Day or night.

3. Standard. The trainee shall operate the torpedo weapon system IAW refs, to include:

   a. Flash-up of system;

   b. Configuration of the Torpedo Status Panel;

   c. Verify weapon's veto panel is enabled;

d. Configuring torpedo weapon system to achieve Barrel Ready condition;

e. Test firing a torpedo in all modes.

4. Time Required.

a. EO = 14 x 50 min periods; and

b. EC = 1 x 20 min period.

5. Instructional Method.

a. Lecture; and

b. Demonstration.

6. References.

a. A01 – A07; and

b. L01 – L03.

7. Training Aids.

a. STG Approved Safety Brief;

b. Flash-up checklist; and

c. Functional check checklist;

8. Test Details.

a. EO 011.02 will be tested with EC 011.02 and PC 011 IAW Annex H.

9. Remarks.

a. Nil.

10. Teaching Points.

a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Demonstrating turn on and off procedures; 2. Explaining each “button” the TSP; 3. Stating the purpose of the toggle switches; 4. Describing modes of operation to include: a. Launch Mode; b. Self-test; c. Maintenance; d. Setup; and e. Training Modes. 5. Reviewing torpedo factory settings.	L	1	
TP2	1. Describe the requirements to illuminate the Barrel Ready Light interlock circuit to include: a. Breech mechanism closed; b. 100psi available; c. Tube door open; d. Flask charged; and e. Switch Actuator Plate open. 2. Explaining and locating VETO system RSB’s for SVTT; 3. Explaining the procedures for firing modes to include: a. Primary via CMS; b. Secondary, locally from the TSP; and c. Emergency.	L	1	

TP3	1. Perform test-firing in all modes, to include: a. Obtain permission to test-fire from Command; b. Ensure areas within firing arcs are clear of personnel and material; c. Ensure safety signs are in place and danger areas are cordoned off; and d. Ensure that any required informative announcements are made prior to firing.			

PO 011 – Maintain Surface Vessel Torpedo Tube

EO 011.03 Repair Torpedo Weapon System

1. Performance. Repair Torpedo Weapon System.

2. Conditions.

   a. Given:

      (1) Supervision;

      (2) References;

      (3) Tools;

      (4) Job aids;

      (5) PPE;

      (6) Assistance; and

      (7) Access to SVTT or training simulator.

   b. Denied:

      (1) Nil.

   c. Environment:

      (1) Classroom;

      (2) HMC ship or SVTT trainer; and

      (3) Day or night.

3. Standard. The trainee shall repair the torpedo weapon system IAW refs, by:

   a. Performing CM on Torpedo Weapon System to include:

      (1) Power faults;

      (2) Pneumatic faults; and

      (3) Electronic faults.

4. Time Required.

   a. EO = 14 x 50 min periods; and

   b. PC = 1 x 20 min period.

5. Instructional Method.

   a. Lecture; and

   b. Demonstration.

6. References.

   a. A01 – A07; and

   b. L01 – L03.

7. Training Aids.

   a. STG Approved Safety Brief;

   b. Flash-up checklist; and

   c. Functional check checklist;

8. Test Details.

   a. EO 011.03 will be tested with EC 011.03.

9. Remarks.

   a. Nil.

10. Teaching Points.

   a. L = Lecture, D = Demonstration, P = Trainee Practice.

	Description	Method	Time	Reference
TP1	1. Conduct CM on torpedo weapon system, to include: a. Explain torpedo weapon system function in detail; b. Identify troubleshooting resources; c. Identify faults to the LRU level; d. Explain impact of faults on system operation; e. Repair or replace affected LRUs; f. Perform functional checks; g. Report equipment status to Command; and h. Complete required technical administration.	L	1	

QUALIFICATION STANDARD ANNEXES

MARPAC RDIMS #1417022
Chap 1 & 2 approved 06 2024
Chap 3 & 4 approved 06 2024

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A01	SEMS	Class Specific Safety and Environment Management System (SEMS)
A02	B-GN-007-RCN/FP-002	CFCD 133 RCN ERT Manual
A03	C-01-000-000/DA-001	NDID Coding System
A04	SSO	Class Specific Ship's Standing Orders (SSOs)
A05	CF H Svcs Gp Instruction 4440-01	Chemical Hazards Surveillance Program
A06	ERT Pack	Class Specific ERT Pack
A07	DAOD 4003-0	Environmental Protection and Stewardship
NAVIGATION DATA DISTRIBUTION SYSTEM		
B01	C-57-858-000/CD	CANAVMOD NDDS (Latest Version)
B02	C-57-858-000/MS-002	LAN INFORMATION FOR HFX CLASS NDDS
INERTIAL NAVIAGATION SYSTEMS		
C01	C-57-794-000/MS-001	Inertial Navigation System MK-49, Operation and Maintenance Manual with Parts List
SHINNADS		
D01	C-57-787-000/MS-001	SHINNADS Dual-M Operation &Maintenance Manual with parts list
SONOBUOY PROCESSING SYSTEM		
E01	C-59-667-000/MS-002	Shipboard Sonobuoy Receiver Assembly, Operation and Maintenance
E02	C-59-765-001/MS-001	Acoustic Signal Processor AN/UYS-503(V)1, Operations and Maintenance, Volume 1
E03	C-59-765-001/MS-002	Acoustic Signal Processor AN/UYS-503(V)1, Drawings and Parts List, Volume 2
E04	C-59-807-000/MS-001	Shipboard Sonobuoy
E05	C-59-667-001/MS-002	Operation & Maintenance InstructionW/IPB - Radio Receiving Set AN/ARR-75
CANTASS		

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
F01	C-69-745-000/MS-001	Description, Operation and Maintenance, Towed Array Handling and Stowage Group,Volume 1
F02	C-69-745-000/MS-Z01	Description, Operation and Maintenance, Towed Array Handling and Stowage Group, Supplement
F03	C-69-745-000/MS-002	Description, Operation and Maintenance, Towed Array Handling and Stowage Group,Volume 2
F04	C-69-745-000/MM-001	Corrective Maintenance, Towed ArrayHandling and Stowage Group volume 2
F05	C-69-746-000/NK-001	PMPT, Towed Array Group
F06	C-69-746-000/NK-002	PMPT, Towed Array Group
F07	C-69-746-000/NK-003	PMPT, Towed Array Group
F08	C-69-746-000/MS-001	Towed Array Group, Description,Operation and Maintenance
F09	C-69-746-B00/TB-002	CANTASS (Tow the line)
F10	C-69-746-B00/TB-00	CANTASS (Tow the line)
F11	C-69-746-B00/TB-003	CANTASS (Tow the line)
F12	C-69-746-000/TB-004	CANTASS (Tow the line)
F13	C-69-748-000/MS-001	CANTASS, Shipboard ElectronicsSubsystem, 2 Books
F14	C-69-748-CC0/MS-001	CANTASS SESS Data Management &Distribution
F15	C-69-748-CEE/MS-001	Un-Interruptible Power Supply
F16	C-69-747-000/MS-001	CANTASS Maintenance Manual Vol. 1
F17	C-69-747-000/MS-002	CANTASS Maintenance Manual Vol. 2
HULL MOUNTED SONAR		
G01	C-69-728-000/MS-001	Operation and Maintenance Manual VIP 200/250
G02	C-69-775-000/MS-001	Control, Sonar Detecting-Ranging Set, C-5575(V)2/SQS- 510(V)4, SP Cabinet

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
G03	C-69-799-AA0/MS-001	Control, Sonar Detecting-Ranging Set, C-5575(V)2/SQS- 510(V4), Sonar Control Cabinet
G04	C-69-799-000/MS-001	Detecting-Ranging Set, Sonar AN/SQS- 510(V)4, 2 Books
G05	C-69-799-C00/MS-001	Group, Sonar OT-5014 (V)3/SQS-510(V) Operating Instructions
G06	C-69-799-000/MB-001	Detecting-Ranging Set, Sonar, AN/SQS-510(V)4, Operating Instruction
TORPEDO DECOY SYSTEM		
H01	C-69-705-000/MS-001	Torpedo Countermeasures Transmitting Set AN/SLQ- 25A(CN) System Operation and Maintenance
H02	C-69-713-M00/NY-001	PM Schedule Interface Control Towed Torpedo Decoy
H03	C-69-713-M00/NY-Z01	PM Schedule Interface Control Towed Torpedo Decoy
SEOSS		
I01	C-58-413-000/MS-002	Operator Manual For SEOSS.SD,Synchro-To-Digital Converter, Model SDC-2060
I02		Shipboard Electo-Optic Surveillance System (SEOSS) AirDryer Maintenance Manual
MAINTAIN COMBAT MANAGEMENT SYSTEM (CMS 330)		
J01	C-60-236-000/MD-001	Equipment Description - Combat Management System (CMS) 330
J02	C-60-236-000/MM-001	First Line Corrective Maintenance Instructions - Volume 1 - Combat Management System (CMS) 330
J03	C-60-236-000/MY-001	Parts Identification List - Combat Management System (CMS) 330

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
J04	C-60-236-000/NK-001	Naval Preventive Maintenance Performance Test - Combat Management System (CMS) 330
J05	C-60-236-000/NY-001	Naval Preventive Maintenance Schedule- Combat Management System (CMS) 330
J06	C-60-239-000/MD-001	Equipment Description - Radar Distribution Unit/Video Switching Unit
J07	C-60-239-000/MM-001	First Line Corrective Maintenance Instructions - Radar Distribution Unit/Video Switching Unit
J08	C-60-239-000/MY-001	Parts Identification List - Radar Distribution Unit/Video Switching Unit
J09	C-60-239-000/NK-001	Naval Preventive Maintenance Performance Test - Radar Distribution Unit/Video Switching Unit
J10	C-60-239-000/NY-001	Naval Preventive Maintenance Schedule - Radar Distribution Unit/Video Switching Unit
J11	4/LZUM1130166	Sea Giraffe AMB H, Operators Training
J12	5/LZUM1130166	Sea Giraffe AMB H, Maintainer Training
TORPEDO HANDLING SYSTEM		
K01	C-69-725-000/MS-001	INSTALLATION, OPERATION, MAINTENANCE AND REPAIR INSTRUCTIONS
K02	C-69-729-000/MN-001	MAINTENANCE MANUEL FOR MARK 46 MOD 5 TORPEDO ILLUSTRATED PARTS BREAKDOWN
K03	NCPM 1001	HALIFAX CLASS MODERNIZATION UNDER WATER WARFARE SURFACE VESSEL TORPEDO TUBES DRILLS AND PROCEDURES MANUAL
K04	C-69-725-F00/MS-001	TORPEDO LOADING TRAY MARK 3 MODS 0 AND 1 - DESCRIPTION, INSTALLATION AND OPERATION APPLICABLE TO HALIFAX CLASS

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
K05	C-75-315-000/MS-000	MAINTENANCE INSTRUCTIONS WITH PARTS LIST - TORPEDO HANDLING EQUIPMENT - MK 24 DOLLY
SURFACE VESSEL TORPEDO TUBES		
L01	C-69-771-000/MS-001	DESCRIPTION, OPERATION, MAINTENANCE AND ILLUSTRATED PARTS BREAKDOWN SURFACE VESSEL TORPEDO TUBE MARK 32 MOD 9
L02	NAVORD 3136-4	Weapons Keys for Ships and Submarines
L03	C-69-771-000/NY-001	NAVAL PREVENTIVE MAINTENANCE SCHEDULE (SONAR) TORPEDO TUBE ASSEMBLY MK32 MOD 9 APPLICABLE TO HALIFAX CLASS

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

JBOS/OSS TASK NO.	TASK ELEMENT	TRG / NO TRG	PO
	NAVIGATION DATA DISTRIBUTION SYSTEM **(Master Task list has it as Nav Switchboard)**		
A1	Initialize a Navigation Distribution System to an operational state		
A2	Install a new or repaired/refurbished/overhauled Navigation Distribution System component		
A3	Monitor a Navigation Distribution System performance		
A4	Operate the Navigation Distribution System (NDS)		
A5	Perform a Navigation Distribution System functional checks		
A6	Perform a Navigation Distribution System PM activity (including a test/trial/inspection)		
A7	Perform post-repair calibration/testing/tuning/alignment of a Navigation Distribution System		
A8	Remove a faulty component from a Navigation Distribution System		
A9	Repair/refurbish/overhaul a faulty component from a Navigation Distribution System		

A10	Bench test a repaired/refurbished/overhauled Navigation Distribution System component		
A11	Calibrate/test/tune/align a newly installed Navigation Distribution System component		
A12	Conduct preliminary fault diagnostic on an Navigation Distribution System (Tech of the Watch)		
A13	Configure a Navigation Distribution System to meet an operational requirement		
A14	Analyze/evaluate/interpret a Navigation Distribution System functional check result		
A15	Analyze/evaluate/interpret a Navigation Distribution System test/trial/inspection result or report		
A16	Configure/reconfigure Navigation Distribution System firmware (e.g. EPROM, DIP switches)		
A17	Diagnose a Navigation Distribution System fault to the Lowest Repairable Unit		
A18	Direct a corrective maintenance/repair activity on a Navigation Distribution System		
A19	Direct a Navigation Distribution System functional check		
A20	Direct a Navigation Distribution System PM activity (including a test/trial/inspection)		
INERTIAL NAVIGATION SYSTEMS			
B1	Initialize an Internal Navigation System to an operational state (e.g. Inertial navigation and gyro compass).		

B2	Monitor an Internal Navigation System performance		
B3	Operate an Internal Navigation System		
B4	Perform an Internal Navigation System functional check		
B5	Analyze/evaluate/interpret an Internal Navigation System functional check result		
B6	Analyze/evaluate/interpret an Internal Navigation System test/trial/inspection result or report		
B7	Perform post-repair calibration/testing/tuning/alignment of an Internal Navigation System		
B8	Remove a faulty component from an Internal Navigation System		
B9	Repair/refurbish/overhaul a faulty component from an Internal Navigation System		
B10	Bench test a repaired/refurbished/overhauled Internal Navigation System component		
B11	Calibrate/test/tune/align a newly installed Internal Navigation System component		
B12	Conduct preliminary fault diagnostic on an Internal Navigation System (Tech of the Watch)		
B13	Configure an Internal Navigation System to meet an operational requirement		
B14	Configure/reconfigure Internal Navigation System firmware (e.g. EPROM, DIP switches)		

B15	Diagnose an Internal Navigation System fault to the Lowest Repairable Unit		
B16	Direct an Internal Navigation System functional check		
B17	Direct an Internal Navigation System PM activity (including a test/trial/inspection)		
SHIPBOARD INTEGRATED NAVIGATION AND DISPLAY SYSTEM			
C1	Initialize a Shipboard Integrated Navigation and Display System (SHINADS) to an operational state		
C2	Corrective Maintenance (CM) procedures of a Shipboard Integrated Navigation and Display System (SHINADS)		
C3	Operating procedures of a Shipboard Integrated Navigation and Display System (SHINADS)		
C4	Preventative Maintenance (PM) procedures (including tests, trials, and inspections) of a Shipboard Integrated Navigation and Display System (SHINADS)		
C5	Characteristics/capabilities/limitations of a Shipboard Integrated Navigation and Display System (SHINADS)		
C6	Install a new or repaired/refurbished/overhauled SHINADS system component		
C7	Monitor a SHINADS system performance		
C8	Operate a SHINADS system		
C9	Perform a SHINADS functional check		

C10	Perform a SHINADS system PM activity (including a test/trial/inspection)		
C11	Perform post-repair calibration/testing/tuning/alignment of a SHINADS system		
C12	Perform post-repair calibration/testing/tuning/alignment of a SHINADS system		
C13	Remove a faulty component from a SHINADS system		
C14	Repair/refurbish/overhaul a faulty component from a SHINADS system		
C15	Bench test a repaired/refurbished/overhauled SHINADS system component		
C16	Calibrate/test/tune/align a newly installed SHINADS system component		
C17	Conduct preliminary fault diagnostic on a SHINADS system (Tech of the Watch)		
C18	Configure/reconfigure SHINADS system firmware (e.g. EPROM, DIP switches)		
C19	Configure/reconfigure SHINADS system software		
C20	Diagnose a SHINADS system fault to the Lowest Repairable Unit		
C21	Direct a corrective maintenance/repair activity on a SHINADS system		
C22	Analyze/evaluate/interpret a SHINADS functional check result		
C23	Analyze/evaluate/interpret a SHINADS system test/trial/inspection result or report		

SONOBUOY PROCESSING SYSTEM			
D1	Corrective Maintenance (CM) procedures of a Sonobouy Processing System		
D2	Operating procedures of a Sonobouy Processing System		
D3	Preventative Maintenance (PM) procedures (including tests, trials, and inspections) of a Sonobouy Processing System		
D4	Characteristics/capabilities/limitations of a Sonobouy Processing System		
D5	Initialize a Sonobouy Processing System to an operational state		
D6	Install a new or repaired/refurbished/overhauled Sonobouy Processing System component		
D7	Operate a Sonobouy Processing System		
D8	Perform a Sonobouy Processing System functional check		
D9	Perform a Sonobouy Processing System PM activity (including a test/trial/inspection)		
D10	Perform post-repair calibration/testing/tuning/alignment of a Sonobouy Processing System		
D11	Remove a faulty component from a Sonobouy Processing System		
D12	Repair/refurbish/overhaul a faulty component from a Sonobouy Processing System		

D13	Bench test a repaired/refurbished/overhauled Sonobouy Processing System component		
D14	Calibrate/test/tune/align a newly installed Sonobouy Processing System component		
D15	Conduct preliminary fault diagnostic on a Sonobouy Processing System (Tech of the Watch)		
D16	Configure a Sonobouy Processing System to meet an operational requirement		
D17	Configure/reconfigure Sonobouy Processing System firmware (e.g. EPROM, DIP switches)		
D18	Configure/reconfigure Sonobouy Processing System software		
D19	Diagnose a Sonobouy Processing System fault to the Lowest Repairable Unit		
D20	Direct a corrective maintenance/repair activity on a Sonobouy Processing System		
D21	Analyze/evaluate/interpret a Sonobouy Processing System functional check result		
D22	Direct a Sonobouy Processing System functional check		
D23	Direct a Sonobouy Processing System PM activity (including a test/trial/inspection)		
D24	Analyze/evaluate/interpret a Sonobouy Processing System test/trial/inspection result or report		
TOWED ARRAY SONAR SYSTEM			

E1	Fill a Towed Array Acoustic Module		
E2	Analyze/evaluate/interpret a TASS functional check result		
E3	Analyze/evaluate/interpret a TASS test/trial/inspection result or report		
E4	Initialize a Towed Array Acoustic Modules system to an operational state		
E5	Initialize a Towed Array Sonar System (TASS) to an operational state (e.g. CANTASS/SUBTASS processing equipment and Array Receiver)		
E6	Perform post-repair calibration/testing/tuning/alignment of a Towed Array Acoustic Modules system		
E7	Initialize an Acoustic Sensor Handling Equipment System to an operational state (e.g. VDS, CANTASS, handling systems).		
E8	Perform post-repair calibration/testing/tuning/alignment of an Acoustic Sensor Handling Equipment System		
E9	Repair/refurbish/overhaul a faulty component from a Towed Array Acoustic Modules system		
E10	Perform the duties of the acoustic sensor handling system Array Handler		
E11	Perform the duties of the acoustic sensor handling system Controller		
E12	Perform the duties of the acoustic sensor handling system Operator		
E13	Perform the duties of the acoustic sensor handling system Safety Observer		

E14	Install a new or repaired/refurbished/overhauled Acoustic Sensor Handling Equipment System component		
E15	Remove a faulty component from a Towed Array Acoustic Modules system		
E16	Direct a corrective maintenance/repair activity on a Towed Array Acoustic Modules system		
E17	Direct a corrective maintenance/repair activity on an Acoustic Sensor Handling Equipment System		
E18	Configure a Towed Array Acoustic Modules system to meet an operational requirement		
E19	Configure an Acoustic Sensor Handling Equipment System to meet an operational requirement		
E20	Direct a Towed Array Acoustic Modules system PM activity (including a test/trial/inspection)		
E21	Direct an Acoustic Sensor Handling Equipment System functional check		
E22	Direct an Acoustic Sensor Handling Equipment System PM activity (including a test/trial/inspection)		
E23	Direct a Towed Array Acoustic Modules system functional check		
E24	Configure/reconfigure Acoustic Sensor Handling Equipment System firmware (e.g. EPROM, DIP switches)		
E25	Diagnose a Towed Array Acoustic Modules system fault to the Lowest Repairable Unit		
E26	Diagnose an Acoustic Sensor Handling Equipment System to the Lowest Repairable Unit		

E27	Configure/reconfigure Sonar System firmware (e.g. EPROM, DIP switches)		
E28	Deploy/recover an acoustic sensor (e.g. VDS, Towed Array)		
E29	Analyze/evaluate/interpret a Towed Array Acoustic Modules system functional check result		
E30	Analyze/evaluate/interpret a Towed Array Acoustic Modules system test/trial/inspection result or report		
E31	Recover Acoustic Sensor in emergency Mode		
E32	Analyze/evaluate/interpret a Towed Array Acoustic Modules system test/trial/inspection result/report result		
E33	Install a new or repaired/refurbished/overhauled TASS component		
E34	Install a new or repaired/refurbished/overhauled Towed Array Acoustic Modules system component		
E35	Repair/refurbish/overhaul a faulty component from an Acoustic Sensor Handling Equipment System		
E36	Bench test a repaired/refurbished/overhauled Acoustic Sensor Handling Equipment System component		
E37	Install/remove a CANTASS cable		
E38	Conduct preliminary fault diagnostic on an Acoustic Sensor Handling Equipment System (Tech of the Watch)		
E39	Conduct preliminary fault diagnostic on a Towed Array Acoustic Modules system (Tech of the Watch)		

E40	Adjust an acoustic sensor cable out		
E41	Calibrate/test/tune/align a newly installed Towed Array Acoustic Modules system component		
E42	Calibrate/test/tune/align a newly installed Acoustic Sensor Handling Equipment System component		
E43	Remove a faulty component from an Acoustic Sensor Handling Equipment System		
E44	Monitor a TASS performance		
E45	Monitor a Towed Array Acoustic Modules system performance		
E46	Bench test a repaired/refurbished/overhauled Towed Array Acoustic Modules system component		
E47	Analyze/evaluate/interpret an Acoustic Sensor Handling Equipment System functional check result		
E48	Monitor an Acoustic Sensor Handling Equipment System performance		
E49	Analyze/evaluate/interpret an Acoustic Sensor Handling Equipment System test/trial/inspection result or report		
E50	Operate a TASS		
E51	Operate an Acoustic Sensor Handling Equipment System		
E52	Perform a TASS functional check		
E53	Perform a TASS PM activity (including a test/trial/inspection)		

E54	Perform a Towed Array Acoustic Modules system functional check		
E55	Perform a Towed Array Acoustic Modules system PM activity (including a test/trial/inspection)		
E56	Perform an Acoustic Sensor Handling Equipment System functional check		
E57	Perform an Acoustic Sensor Handling Equipment System PM activity (including a test/trial/inspection)		
E58	Perform an acoustic sensor handling system post- recovery checks/routine		
E59	Perform an acoustic sensor handling system pre- launch checks/routine		
E60	Perform post-repair calibration/testing/tuning/alignment of a TASS		
E61	Remove a faulty component from a TASS		
E62	Repair/refurbish/overhaul a faulty component from a TASS		
E63	Bench test a repaired/refurbished/overhauled TASS component		
E64	Calibrate/test/tune/align a newly installed TASS component		
E65	Conduct preliminary fault diagnostic on a TASS (Tech of the Watch)		
E66	Configure a TASS to meet an operational requirement		
E67	Configure/reconfigure TASS firmware (e.g. EPROM, DIP switches)		
E68	Configure/reconfigure TASS software		

E69	Diagnose a TASS fault to the Lowest Repairable Unit		
E70	Direct a corrective maintenance/repair activity on a TASS		
E71	Direct a TASS functional check		
E72	Direct a TASS PM activity (including a test/trial/inspection)		
HULL MOUNTED SONAR SYSTEM			
F1	Initialize a Sonar System to an operational state. (e.g. 510, Sonar 2008 NG, Sonar 2009 Variant AAA, Side Scan Sonar)		
F2	Install a new or repaired/refurbished/overhauled Sonar System component		
F3	Monitor a Sonar System performance		
F4	Operate a Sonar System		
F5	Perform a Sonar System functional check		
F6	Perform a Sonar System PM activity (including a test/trial/inspection)		
F7	Perform post-repair calibration/testing/tuning/alignment of a Sonar System		
F8	Remove a faulty component from a Sonar System		
F9	Conduct preliminary fault diagnostic on a Sonar System (Tech of the Watch)		

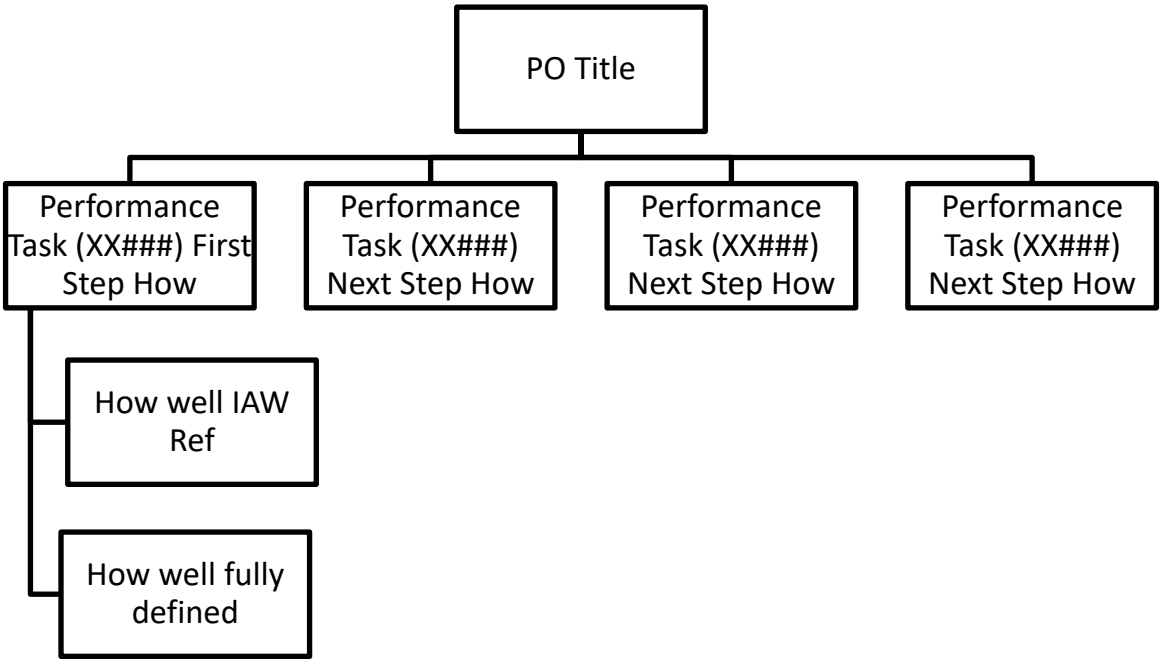
F10	Configure a Sonar System to meet an operational requirement		
F11	Diagnose a Sonar System fault to the Lowest Repairable Unit		
F12	Calibrate/test/tune/align a newly installed Sonar System component		
TORPEDO DECOY SYSTEMS			
G1	Initialize a Torpedo Decoy System to an operational state		
G2	Install a new or repaired/refurbished/overhauled Torpedo Decoy System component		
G3	Install a NIXIE towed body		
G4	Launch Torpedo Decoy System in emergency mode		
G5	Monitor a Torpedo Decoy System performance		
G6	Operate a Torpedo Decoy System		
G7	Perform a Torpedo Decoy System functional check		
G8	Perform a Torpedo Decoy System pre-launch check/routine		
G9	Perform a Torpedo Decoy System post-recovery check/routine		
G10	Perform a Torpedo Decoy System PM activity (including a test/trial/inspection)		
G11	Perform post-repair calibration/testing/tuning/alignment of a Torpedo Decoy System		

G12	Perform the duties of the Cable Handler		
G13	Perform the duties of the Winch Operator		
G14	Remove a faulty component from a Torpedo Decoy System		
G15	Remove a NIXIE towed body		
G16	Adjust a Torpedo Decoy System cable out		
G17	Calibrate/test/tune/align a newly installed Torpedo Decoy System component		
G18	Conduct preliminary fault diagnostic on a Torpedo Decoy System (Tech of the Watch)		
G19	Configure a Torpedo Decoy System to meet an operational requirement		
G20	Diagnose a Torpedo Decoy System fault to the Lowest Repairable Unit		
SHIPBOARD ELECTRO OPTIC SURVEILLANCE SYSTEM			
H1	Initialize an Shipboard Electro Optic Surveillance System		
H2	Install a new or repaired/refurbished/overhauled Shipboard Electro Optic Surveillance System component		
H3	Monitor a Shipboard Electro Optic Surveillance System		
H4	Operate a Shipboard Electro Optic Surveillance System		

H5	Perform an Shipboard Electro Optic Surveillance System functional checks		
H6	Perform an Shipboard Electro Optic Surveillance System PM activity (including a test/trial/inspection)		
H7	Analyze/evaluate/interpret an Shipboard Electro Optic Surveillance System		
H8	Perform post-repair calibration/testing/tuning/alignment of an Shipboard Electro Optic Surveillance System		
H9	Remove a faulty component from an Shipboard Electro Optic Surveillance System		
H10	Repair/refurbish/overhaul a faulty component from an Shipboard Electro Optic Surveillance System		
H11	Bench test a repaired/refurbished/overhauled Shipboard Electro Optic Surveillance System		
H12	Calibrate/test/tune/align a newly installed Shipboard Electro Optic Surveillance System		
H13	Analyze/evaluate/interpret a Shipboard Electro Optic Surveillance System functional checks results		
H14	Conduct preliminary fault diagnostic on an Shipboard Electro Optic Surveillance System		
H15	Configure a Shipboard Electro Optic Surveillance System to meet operational requirements		
H16	Diagnose an Surveillance System to meet operational to the Lowest Repairable Unit		
H17	Direct an Shipboard Electro Optic Surveillance System functional check		

H18	Direct an Shipboard Electro Optic Surveillance System PM activity (including a test/trial/inspection)		
COMBAT MANAGEMENT SYSTEM			
J1	Initialize a Command and Control System (CCS) to an operational state		
J2	Monitor a CCS performance		
J3	Install a new or repaired/refurbished/overhauled CCS component		
J4	Operate a CCS		
J5	Perform a CCS functional check		
J6	Calibrate/test/tune/align a newly installed CCS component		
J7	Configure a CCS to meet an operational requirement		
J8	Diagnose a CCS fault to the Lowest Repairable Unit		
J9	Perform post-repair calibration/testing/tuning/alignment of a CCS		

ANNEX C – QS SCALAR New requirement under new format.



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ANNEX D – RECORD OF DECISIONS QSWB

Dates:

28 June – 9 July 2010

Location:

CFB Esquimalt C&PO's Mess, Pacific Station

Board Members:

SURNAME	INITIALS	RANK	MOSID	UNIT
Bell	GD	CPO1	00119	CANFLTPAC HQ / Chair
Giroux	C	Lt(N)	00211	CFFSE / TDO
Magee	MG	CPO1	00119	DMTE Rep
Stonier	PR	CPO1	00119	D Mar Pers Rep
Cahoon	WD	CPO2	00119	CFFSE / SME
Pollard	LJ	PO1	00017	CFFSE / SME
Gormley	GP	PO2	00116	HMCS VICTORIA / SME
Jensen	NV	PO2	00116	HMCS REGINA/ SME
Whittle	GA	PO2	00116	CFNES / SME
Schmidt	JD	MS	00017	FMF Cape Breton / SME
Fabella	MNS	OS	00299	CFFSE / Secretary

Board Mandate:

1. The mandate of the Weapons Engineering Tech Sonar System Maintainer Qualification Standard and Plan Writing Board (QSPWB) was to create the Qualification Standard and Plan (QSP) for the W Eng Tech Sonar System Maintainer formal training that results in properly qualified personnel capable of carrying out duties commensurate with their rank and MOSID requirements.

Discussions / Decisions:

3. Presentations on Current Evolution of Occupation. When the QSPWB convened, the representative from DMTE, CPO1 Magee, briefed Board members on the direction the Weapons Engineering occupation is headed and what the previous QSPWBs had developed for the Jr W Eng Tech, the W Eng Tech, the W Eng Radar System Maintainer, and the W Eng Communications System Maintainer occupations.

4. System Maintainer Philosophy – Detection Equipment to Weapons Delivery. The D Mar Pers representative, CPO1 Stonier, reiterated to the Board members that the difference between the W Eng Tech and the W Eng System Maintainer was the W Eng Tech would maintain a single piece of equipment whereas the W Eng System Maintainer would be responsible to maintain a system that went from the “Detection equipment to the weapon delivery equipment”, for example from the sonar system to the torpedo.

5. Sequence of Equipment Training. The Board members reviewed the equipment responsibilities for the W Eng Sonar System Maintainer and recommended the following training sequence:

a. Doppler Speed Log (DSL);
b. Mk49 SINS;
c. SHINNADS;
d. Sonobuoy Processing System;
e. CANTASS;
f. AN/SQS-510;
g. NIXIE;
h. SEOSS; and
i. CCS.

6. The Board commenced chapter 4 by utilizing the present lesson plans from the QL3 and QL5 NE Tech A courses for the Doppler Speed Log, Mk49 SINS and SHINNADS.

7. The TDO, Lt(N) Giroux, suggested that a statement be added to the QSP indicating that when a PC is administered where there weren’t ECs prior to the PC, that the Instructor is to provide a debrief to the student following each practice fault.

7. The TDO, Lt(N) Giroux, suggested that the verb “Observe” be replaced with “Monitor”. The Board agreed and changed instituted the changes throughout the QSP
.

8. The Board commenced the EOs for CANTASS. After much discussion it was decided to break the CANTASS into essentially “wet” and “dry” ends of the system. It was decided to have 5 EOs for CANTASS, 2 that would have enabling checks – the H&SG/Array and the SESS.

9. The Board received the “suggested” statement from the TDO; it was decided by the Board that some “word smithing” would be required to shorten the statement and make it straight forward.

11. The Board completed the EOs for CANTASS. It was decided by the Board that upon completion of the EC for EO 005.05, there would be two “system” practice faults for each student. The practice system faults would be 90 minutes/fault vice the standard 45 minutes as the system is so diverse.

12. Upon completion of the day’s work; the representative from D Mar Pers, CPO1 Stonier, thanked all members of the Board for their contributions to the Weapons Engineering Sonar occupation and encouraged them to maintain their diligence during the final week of the Board.

13. The Board commenced and completed the EOs for the AN/SQS-510 and the NIXIE. During the discussions for the AN/SQS-510 it was decided that because the SP in the 510 was so similar to the HSSP in CANTASS, not as much time would be required to detail the function of the similar CCAs. However if the sequencing of training were such that the 510 was trained before CANTASS, then the emphasis would be put on the CCAs in the 510 SP and when the CANTASS HSSP was trained then the differences could be applied then and the CANTASS HSSP training could be reduced. As the Torpedo Band/De-Band is presently taught on the NW Tech QL3, it was decided to remove it as a PO for this course and re- integrate it into the Jr W Eng Tech Mod 2 and Mod 3.

<u>Guidance to Instructional Designers:</u>
<u>Follow-up Action:</u> DMTE to submit QSP for approval.
Other Considerations (if applicable)

TRAINING PLAN ANNEXES

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. Facilities/Trainers

SER	DESCRIPTION		QTY	REMARKS
a.				EO001.02 and 009.11 Security Classification required
b.				
c.				
NOTES:				

3. Ammunition

SER	LOG GUIDE	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

4. Weapons

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				

c.	XXX-XX-XXX-XXXX				
NOTES:					

5. Specialized/Technical Equipment

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

6. Transport/Vehicles

SER		NSN	DESCRIPTION		QTY	REMARKS
a.		XXX-XX-XXX-XXXX				
b.		XXX-XX-XXX-XXXX				
c.		XXX-XX-XXX-XXXX				
	NOTES:					

7. Communication Equipment/Frequencies

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				

c.	XXX-XX-XXX-XXXX				
NOTES:					

8. Training Aids

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

9. POL Products

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

10. General Purpose Stores

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

11. Stationery

SER	DESCRIPTION	QTY	REMARKS

a.				
b.				
c.				

NOTES:		

12. Personal Equipment

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION New requirement under new format values in table not accurate for Sonar example only.

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: EXAMPLE		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	14	$1,000.0		$14,000	$14,000	$70,000.00
R&Q	14	$36.0	20	$10,080	$10,080	$50,400.00
Incidentals	14	$17.5	20	$4,900	$4,900	$24,500.00
Subtotal				**$28,980**	**$28,980**	**$144,900.00**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	16	$50.0	1	$800	$800	$4,000.00
Classroom (square meters)	30	$13.00	15	…	…	…
Computer requirements	16	$1,000.0	1	$16,000	$0	$16,000.00
IT Equip (Projector, Smart board etc.)	1	$6,000.0	1	$6,000	$0	$6,000.00
DWAN connection (Set up)	1	$5,000.0	1	$5,000	$0	$5,000.00
Link Toolkit (on DVD)	15	$1.0	1	$15	$15	$75
Instructional Materials (Printed / Handouts etc.)	15	$30.0	1	$450	$450	$2,250.00
Subtotal				**$28,265**	**$1,265**	**$33,325.00**

PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2	2	$223.0	20	$8,920	$8,920	$44,600.00
Incremental Civilian Instructor		XXX				
		subtotal		**$8,920**	**$8,920**	**$44,600.00**
	Total Estimated Costs			**$66,165**	**$39,165**	**$222,825**

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

Allocation of ATT	**Minutes**
Actual Training Time	560 periods
ATT Sub-total	
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	
Training Administrative Time (TAT)	
Training Travel Time (TTT)	
Physical Fitness (PT)	
Sub-total	
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**90 days**

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PO/EO	OBJECTIVE	EO	EC	PC	PO
PO 001	Maintain Navigation Distribution System	44		14.0	58.0
EO 001.01	Operate a Navigation Distribution System	17.0			
EO 001.02	Repair a Navigation Distribution System Fault	27.0		14	
PO 002	Maintain Inertial Navigation Systems	35.5		8.0	43.5
EO 002.01	Describe the Inertial Navigation Systems	14.5			
EO 002.02	Initialize the Inertial Navigation Systems	5.0			
EO 002.03	Repair the Inertial Navigation Systems	16.0		8.0	
PO 003	Maintain the Shipboard Integrated Navigation System	32.0		8.0	40.0
EO 003.01	Describe the SHINNADS Dual Monitor	13.0			
EO 003.02	Initialize the SHINNADS Dual Monitor	3.0			
EO 003.03	Repair the SHINNADS Dual Monitor	16.0		8.0	
PO 004	Maintain the Sonobuoy Processing System	25.0		8.0	33.0
EO 004.01	Describe the Sonobuoy Processing System	6.0			
EO 004.02	Initialize the Sonobuoy Processing System	3.0			
EO 004.03	Repair the Sonobuoy Processing System	16.0		8.0	
PO 005	Maintain Towed Array Sonar System	187	27.0	19.0	233
EO 005.01	Describe the Canadian Towed Array Sonar System	15.0			
EO 005.02	Initialize the Towed Array Sonar System	6.0			
EO 005.03	Repair the Array/Handling and Stowage Group	74	17.0		
EO 005.04	Repair the Array Receiver	6.0			
EO 005.05	Repair the Shipboard Electronic Sub- System	86.0	10.0	19.0	
PO 006	Maintain Hull Mounted Sonar	70.5	2.0	1.0	73.5
EO 006.01	Describe the AN/SQS-510 Sonar	15.0			
EO 006.02	Initialize the Hull Mounted Sonar	3.0			
EO 006.03	Repair the AN/SQS-510 Transmitter Group	13.5	1.0		
EO 006.04	Repair the Sonar Operating Console	6.5			
EO 006.05	Repair the Sonar Control and Processing Group	32.5	1.0	1.0	

PO 007	Maintain Torpedo Decoy System	28.0		8.0	36.0
EO 007.01	Describe the AN/SLQ-25A	15.0			
EO 007.02	Initialize the Torpedo Decoy System	1.0			
EO 007.03	Repair the Torpedo Decoy System	12.0		8.0	
PO 008	Maintain Shipboard Electro Optic Surveillance System	31.0		10.0	41.0
EO 008.01	Describe the Shipboard Electro Optic Surveillance System	14.0			
EO 008.02	Initialize the Shipboard Electro Optic Surveillance System	5.0			
EO 008.03	Repair the Shipboard Electro Optic Surveillance System	12.0		10.0	
PO 009	Maintain Combat Management System 330	30		5	35
EO 009.01	Describe the CMS 330	10			
EO 009.02	Initialize the CMS 330	10			
EO 009.03	Repair the CMS 330	10		5	
PO 010	Utilize DRMIS Portal as a Maintenance Supervisor	5		1	6

OPTIMUM TRAINING SCHEDULE **NEW requirement under new format**

ANNEX H – ASSESSMENT PLAN

EC/PC	Scope Covered	Purpose	How	When	Resources	Pass Mark	Impact
PC 001	EO 001.03	To evaluate the Trainee's knowledge to maintain the NDDS_	Practical	O/C of EO 001.02	NDDS Publications & Stationary	60%	Re-attempt TRB
PC002	EO 002.01 thru EO 002.03	To evaluate the Trainee's ability to Maintain the Inertial Navigation System	Practical	O/C EO 002.03	Inertial Navigation System	60%	Re-attempt TRB
PC 003	EO 003.01 Thru EO 003.03	To evaluate the Trainee's ability to Maintain Shipboard Integrated Navigation System	Practical	O/C EO 003.03	Shipboard Integrated Navigation System	60%	Re-attempt TRB

PC 004	EO 004.01 Thru EO 004.03	To evaluate the Trainee's ability to maintain SPS	Practical	O/C of EO 004.03	SPS	60%	Re-attempt TRB
EC 005.03	EO 005.03	To evaluate the Trainee's ability to repair CANTASS	Practical	O/C EO 005.03	CANTASS	60%	Re-attempt TRB
EC 005.05	EO 005.05	To evaluate the Trainee's ability to repair SESS	Practical	O/C of EO 005.05	Shipboard Electronic Sub-System	60%	Re-attempt TRB
PC 005	EO 005.01 Thru EO 005.05	To evaluate the Trainee's ability to maintain CANTASS	Practical	O/C of EO 005.05	CANTASS	60%	Re-attempt TRB
EC 006.03	EO 006.03	To evaluate the Trainee's ability to repair the Hull Mounted Sonar	Practical	O/C EO 006.03	Hull Mounted Sonar system	60%	Re-attempt TRB

EC 006.05	EO 006.05	To evaluate the Trainee's ability to repair Sonar Control and Processing Group	Practical	O/C 006.05	Hull Mounted Sonar system	60%	Re-attempt TRB
PC 006	EO 006.01 Thru EO 006.05	To evaluate the Trainee's ability to maintain the Hull Mounted Sonar System	Practical	O/C 006.05	Hull Mounted Sonar system	60%	Re-attempt TRB
PC 007	EO 007.01 Thru EO 007.03	To evaluate the Trainee's ability to maintain NIXIE	Practical	O/C 007.03	NIXIE	60%	Re-attempt TRB
PC 008	EO 008.01 Thru EO 008.03	To evaluate the Trainee's ability to maintain SEOSS	Practical	O/C 008.03	SEOSS	60%	Re-attempt TRB

PC 009	EO 009.01 Thru EO 009.02	To evaluate the Trainee's ability to maintain CMS330	Practical	O/C EO 009.03	CMS 330	60%	Re-attempt TRB

ANNEX I – ASSESSMENT CHECKLISTS

From old QSP

PC/EC EVALUATION SHEET			
Equipment:		**Name:**	
PO Number:	**Start:**	**Stop:**	**Total:**
Course:			
Note: When allocating a score use only whole numbers. Circle the appropriate number in column			

Initialization (10%)		
Score	**Justification**	**Remarks**
1 - 2	Did not follow correct flash up procedure. Flash up faults not recognized	
3 - 4	Followed procedure but flash up faults not recognized	
5 - 6	Followed flash up procedure. After some hesitation, corrected flash up faults	
7 - 8	Followed Flash Up procedure. Recognized and corrected flash up faults	
9 - 10	Confidently Flashed Up, efficiently corrected flash up faults.	

Use of Test Equipment (10%)		
Score	**Justification**	**Remarks**

1 - 2	Did Not select appropriate TE for Fault/ Did not demonstrate correct TE operating procedures	
3 - 4	Hesitant with selection of TE/Unsure of TE set up	
5 - 6	Selected correct TE/ Demonstrate weak operating procedures	
7 - 8	Select Correct TE/Properly utilized TE	
9 - 10	Selected Correct TE considering best options/Demonstrated thorough knowledge of operation	
Fault Finding Procedures (30%)		
Score	**Justification**	**Remarks**
1 - 2	Hesitant to begin Fault Finding./Did not recognize fault indications	
3 - 4	Did not correctly interpret Fault Indications; BITE; Test Equipment Readings; Took the wrong path and did not recover.	

5 - 6	Minor inconsistencies in interpreting fault indications. May have taken wrong path but recovered. Employed logical faultfinding.	
7 - 8	Recognized and Correctly interpreted fault indications. Employed Logical Faultfinding.	
9 - 10	Efficiently interpreted Fault Indications and evaluated all symptoms. Found fault	
	with minimum number of steps.	
System Knowledge (30%)		
Score	**Justification**	**Remarks**
1 - 2	System knowledge did not support faultfinding. Could not answer questions from instructor. Could not properly use supporting documents	
3 - 4	System knowledge weak. Could not answer all questions. Struggled with documentation to support faultfinding. Could not fully explain fault	
5 - 6	System knowledge weak in some areas. Answered most questions from instructor and gave general explanations regarding the fault.	

7 - 8	Good system knowledge. Use of supporting documentation aided in fault finding. Explained fault clearly.	
9 - 10	Detailed system knowledge. Used documentation efficiently to find fault. Gave detailed explanation of fault	
Timing (10%)		
Score	**Justification**	**Remarks**
10	Found Fault in allotted time	
8	Found Fault with an extra	
	15 minutes	
6	Found Fault with an extra 20 Minutes	
4	Found Fault with an extra 25 Minutes	
2	Found Fault with an extra 30 Minutes	
Find the Fault (10%)		
Score	**Justification**	**Remarks**
10	Fault Corrected	

Score	**Weighting**	**Total**
Initialization	**X 1**	**%**

Test Equipment	**X 1**	**%**
Fault Finding	**X 3**	**%**
System Knowledge	**X 3**	**%**
Timing	**X 1**	**%**
Finding the Fault	**X 1**	**%**
	TOTAL	**%**

Assessor Signature:	**Trainee Signature:**

PART I - ADMINISTRATIVE INSTRUCTIONS

1. This is a generic list for administering a PC/EC.

2. The Assessor shall:

a. ensure all publications, test equipment and equipment are serviceable or Trainee informed of deficiencies;

b. ensure the Trainees observe all safety precautions;

c. test Trainees individually;

d. supervise and provide physical assistance as required;

e. provide a pre and post assessment waiting area; and

f. review Annex I with the Trainee.

3. If, after allotted time, the Trainee is showing sufficient progress the Assessor may allot 15 additional minutes to complete the PC/EC with the loss of two points. For each additional 5 minutes after the initial 15 minutes will result in two points being lost. The maximum extra time granted will be 30 minutes. If the student does not find the fault after the additional 30 minutes, the instructor will stop the PC/EC and calculate the final mark.

4. The Assessor is to ensure that the best use of time allotted for PC's/EC's is made by facilitating the PC/EC process. For example:

a. if the Trainee is about to perform a logical but time consuming procedure the Assessor should supply the results and have the Trainee resume without penalty to the student however, if the action is not justified by the Trainee it is considered incorrect procedure; or

b. after a maximum of 10 minutes of inactivity the Assessor shall provide assistance.

5. Pass mark is 60%.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. You are given the following:

a. references;

b. course notes;

c. job sheets;

d. work sheets;

e. projects; and

f. equipment and/or MPT

2. IAW the references and PC/EC list, you shall:

a. power up;

b. conduct system functional checks;

c. load software or crypto as required;

d. localize fault to LRU;

e. repair;

f. verify operational readiness;
g. report status to assessor; and

h. power down.

3. Prior to performing a test or measurement you shall inform the Assessor of your intention.

4. IAW the references and PC/EC List, you shall complete the PC/EC:

a. within allotted time IAW Chapter 4;

b. Individually;

c. with supervision; and

d. physical assistance as required.

5. IAW the references, you shall observe all safety precautions. A safety infraction or violation will result in instructor intervention and the instructor may terminate PC/EC based on the severity of the violation.

6. Pass mark is 60%.

7. Do you have any questions?

Trainee Sign: _____Date:_____

Instructor Sign:_____ Date:_____

New format

PERFORMANCE/ENABLING CHECK

PC/EC Time: ### mins (concurrent/per trainee) + ## min debrief **ASSESSMENT ACTIVITY**: (Brief description of the assessment activity that students will do IOT prove they have met the Standard of the PO/EO). **Pass/fail criteria**: (identify success criteria) **ASSESSMENT RESOURCES:** • (Bulleted list of everything given to trainees for PC/EC) **IMPACT OF FAILURE:** Remedial Training / Re-test

INVIGILATOR NOTES:

(Course Name) – PC/EC XXX.XX Assessment Checklist			
Trainee Name:	**Assessor Name:**	**Date:**	**Session:**

1. PASS / FAIL CRITERIA: Identify success criteria.

2. IMPACT OF FAILURE: Remedial training, re-test.

ITEM	**PASS**	**FAIL**	**COMMENTS**
1. Criteria			
(Success criteria taken from PO/EO standard)			
2.			
ADDITIONAL COMMENTS			
STRENGTHS:	**AREAS FOR DEVELOPMENT:**		
THE TRAINEE HAS RECEIVED A DEBRIEF AND SEEN THE CONTENT OF THIS ASSESSMENT			
OVERALL ASSESSMENT (circle one)		**FAIL**	**PASS**
Assessor	Name:	Date:	Signature:

ITEM		PASS	FAIL	COMMENTS
Trainee	Name:	Date:	Signature:	

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ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

Version	Revision Date	Remarks
OJPR Version 1.0	DD / MM/ YYYY	New OJPR

Name and Initials	Rank	Service Number

Date Training Started	DD MMM YYYY
Date of Completion	DD MMM YYYY

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

PROGRESS REVIEW

5. Progress review shall be carried out quarterly by the Divisional Officer and Section Head (NCMs) or Training Officer (Officers).

Example OJPR record:

PO/EO	Description	Remarks	Rank/Name/ Signature	Date	OPI and Unit
PC/EC XXX.XX				DD/MM/YYYY	

OJPR Reviewed By OJT Supervisor (Rank, Position, Name)	Signature	Date
Section Head (NCMs) or HOD (Officers)		

TRAINING COMPLETION CERTIFICATE

Date training completed:	
Div Officer (NCMs) or CO (Officers)	Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed the OJT to the standards stated in the Name of QSP.

TE Cmdt/CO	Date

ANNEX K – TT PLAN

EO	COA 1 TEL Delivery	COA 2 Interim Delivery	**Impact if TEL not implemented** - Realized benefits - E/W Coast Delta

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ANNEX L – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title				
Location of Board				
Dates				
Board Composition				
	Surname	Initials	Rank	Unit
Co-Chair (content)				
Co-Chair (process)				
NFS				
SME				
SME				
SME				
SME				
SME/Secretary				
Advisor				

Day #	Details	Action
	The following information on target population was compiled during the initial steps of the TPWB to give context for design and development: (rank, experience, entry level education, etc.).	
1		
2		

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Amendment Form, thus ensuring the QSP has the most relevant information.

Version:	**X.X**
Date:	**Who Initiated Change:**
Requirement for the Amendment:	
Description of Changes:	
Change Tracking number:	

Trg Mgr:

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.

ACRONYM / ABBREVIATION	TERM/DEFINITION
BITE	Built In Test Equipment
CCS	Command and Control Systems
CSAT	Combat System Alignment Test
CSIT	Combat System Integration Test
CSOT	Combat System Operability Test
DWAN	Defence Wide Area Network
LAN	Local Area Network
LRU	Lowest Replaceable Unit
MPT	Maintenance Procedural Trainer - Can be any form of computer based training, table top faults, or any other form of fault finding practice necessary.
OFL	Offline BITE
POFA	Programmed Operational Functional Analysis
SHIPLAN	Ship Local Area Network
UPS	Uninterruptable Power Supply